

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 1

**Title: Glycolysis and pentose phosphate pathway and their
role in metabolism**

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Required Prereading: Glycolysis (Textbook reading assignment)

- **Before DM Lecture 01**

- Lippincott's Biochemistry 9th Edition

 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>

- **Chapter 8: Introduction to Metabolism and Glycolysis; Sections: Glycolysis overview; Glucose transport into cells; Glycolysis reactions; Hormonal regulation; Alternate fates of pyruvate; Chapter concept map: Fig: 8.25**

- Lippincott's Biochemistry Chap 8 (Concept map Fig 8.25) Glycolysis [Introduction to Metabolism and Glycolysis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

SOM. MK.I. BPM2.3. DM.2.BCHM.1084	Define glycolysis and explain its role in the generation of metabolic energy.
SOM. MK.I. BPM2.3. DM.2.BCHM.1085	List the reactions and enzymes that convert glucose into pyruvate (emphasize key enzymes and irreversible reactions)
SOM. MK.I. BPM2.3. DM.2.BCHM.1086	Describe substrate level phosphorylation and identify those reactions in glycolytic pathway
SOM. MK.I. BPM2.3. DM.2.BCHM.1087	Explain significance of lactate production in anaerobic glycolysis. Explain the further fate of lactate formed in RBC and muscle and highlight importance of the Cori cycle.
SOM. MK.I. BPM2.3. DM.2.BCHM.1088	Outline the regulation of glycolysis indicating the regulatory enzymes. Highlight the role of AMP, ATP on glycolysis.

Objectives: Glycolysis and Pentose phosphate pathway – Role in metabolism

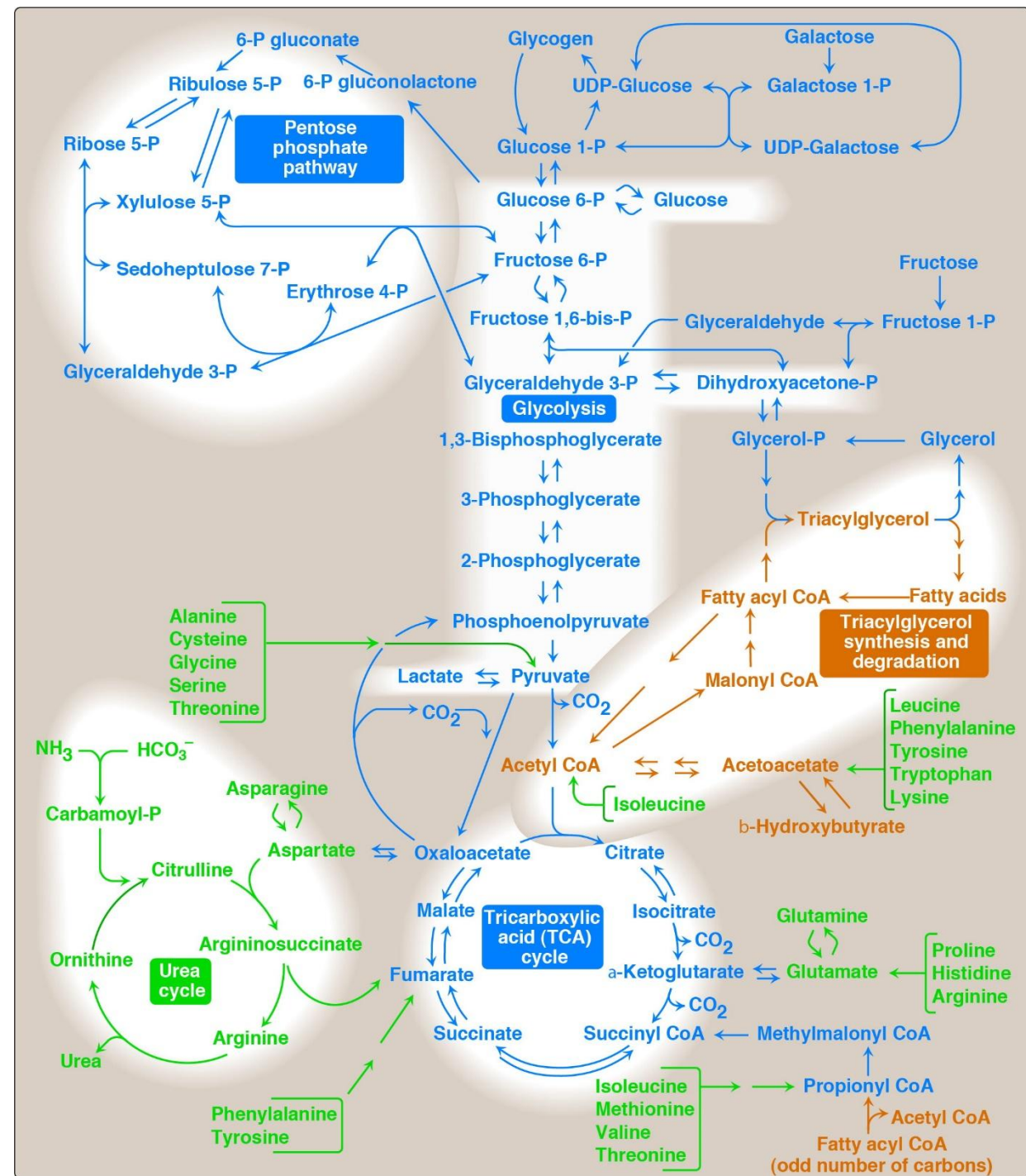
SOM. MK.I. BPM2.3. DM.2.BCHM.1089	Review the roles of glycolysis and pentose phosphate pathway in metabolism
SOM. MK.I. BPM2.3. DM.2.BCHM.1090	Compare and contrast the action of glucokinase and hexokinase (with the help of a graph)
SOM. MK.I. BPM2.3. DM.2.BCHM.1091	Describe the regulation of glycolysis indicating the regulatory enzymes.
SOM. MK.I. BPM2.3. DM.2.BCHM.1092	Appraise the role of AMP, ATP and fructose 2,6 bisphosphate on glycolysis.
SOM. MK.I. BPM2.3. DM.2.BCHM.1093	Highlight the regulation of pyruvate kinase.
SOM. MK.I. BPM2.3. DM.2.BCHM.1094	Explain the role of glucokinase in regulating blood glucose levels. Predict the effect of mutation in pancreatic glucokinase (MODY-2) on blood glucose levels.
SOM. MK.I. BPM2.3. DM.2.BCHM.1095	Outline the function of glycolysis in specific tissues including liver, brain, muscle, eye, tumor cells.
SOM. MK.I. BPM2.3. DM.2.BCHM.1096	Analyze the effect of arsenic on glycolytic enzymes
SOM. MK.I. BPM2.3. DM.2.BCHM.1097	Identify the biochemical mechanisms involved in the occurrence of lactic acidosis in states of poor tissue perfusion/ oxygenation and metabolic states where aerobic metabolism is hindered.
SOM. MK.I. BPM2.3. DM.2.BCHM.1098	Explain the role of NADPH generated by the pentose phosphate pathway. Highlight its use in fatty acid and cholesterol synthesis

Recommended Reading

- Lippincott's Biochemistry 9th Edition
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
 - Chapter 8: Introduction to metabolism and glycolysis; Chapter 8: Introduction to Metabolism and Glycolysis; Sections: Glycolysis overview; Glucose transport into cells; Glycolysis reactions; Hormonal regulation; Alternate fates of pyruvate; Chapter concept map: Fig: 8.25
- Lippincott's Biochemistry Chap 8 (Concept map Fig 8.25) Glycolysis [Introduction to Metabolism and Glycolysis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
Sections: Glycolysis overview; Glucose transport into cells; Glycolysis reactions; Hormonal regulation; Alternate fates of pyruvate; Chapter concept map: Fig: 8.25
- Lippincott's Biochemistry (Pentose phosphate pathway) Chapter 13; Concept map Fig 13.14; [Pentose Phosphate Pathway and Nicotinamide Adenine Dinucleotide Phosphate | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#) : Chapter summary: Fig 13.14
- Lippincott's Biochemistry **Fig 28.29** (Vitamins reference) [Micronutrients: Vitamins | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- **Lippincott's Biochemistry Questions: 8.1 -8.6**

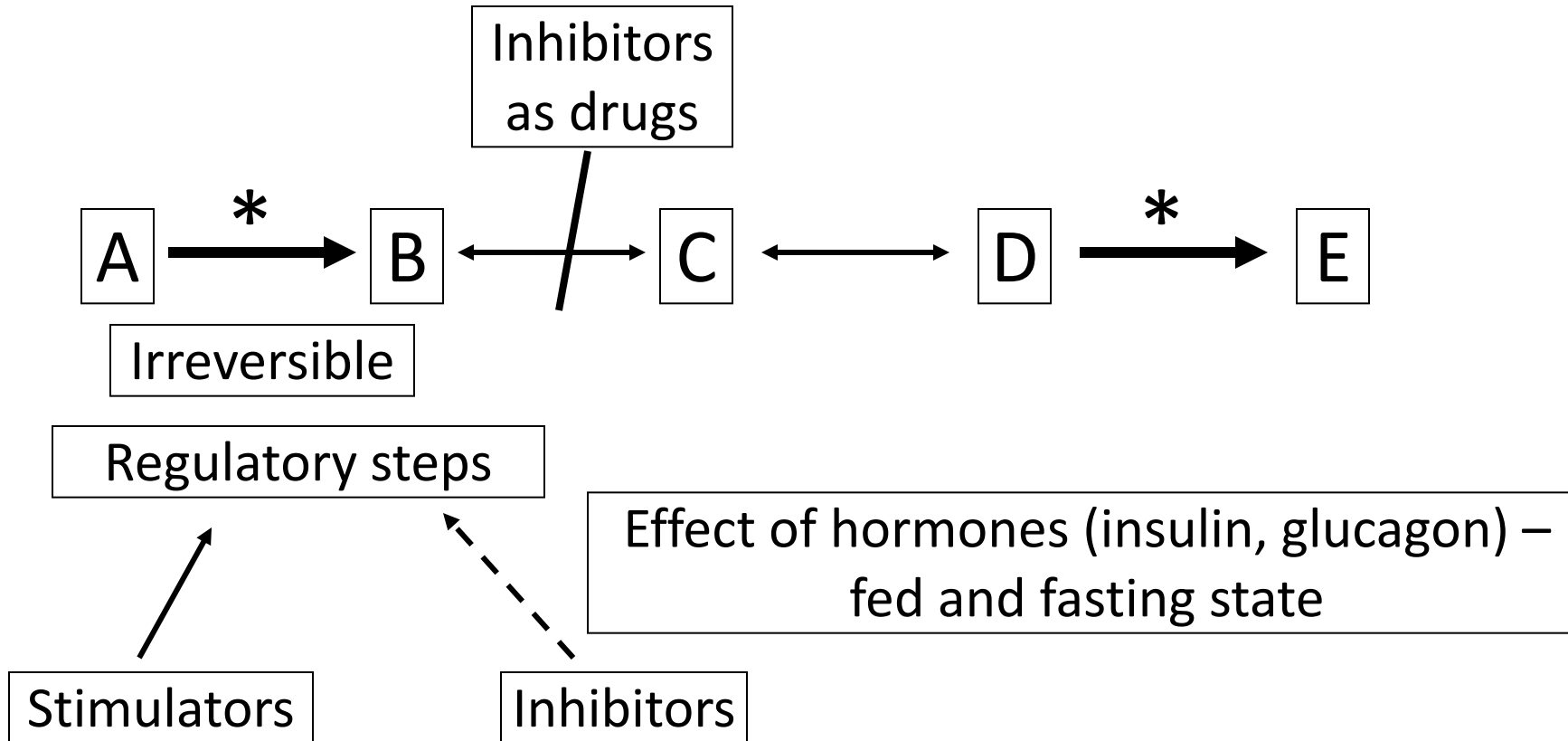
Intermediary metabolism (Metabolic map)

DLA: Overview of metabolism in the fed state and during fasting

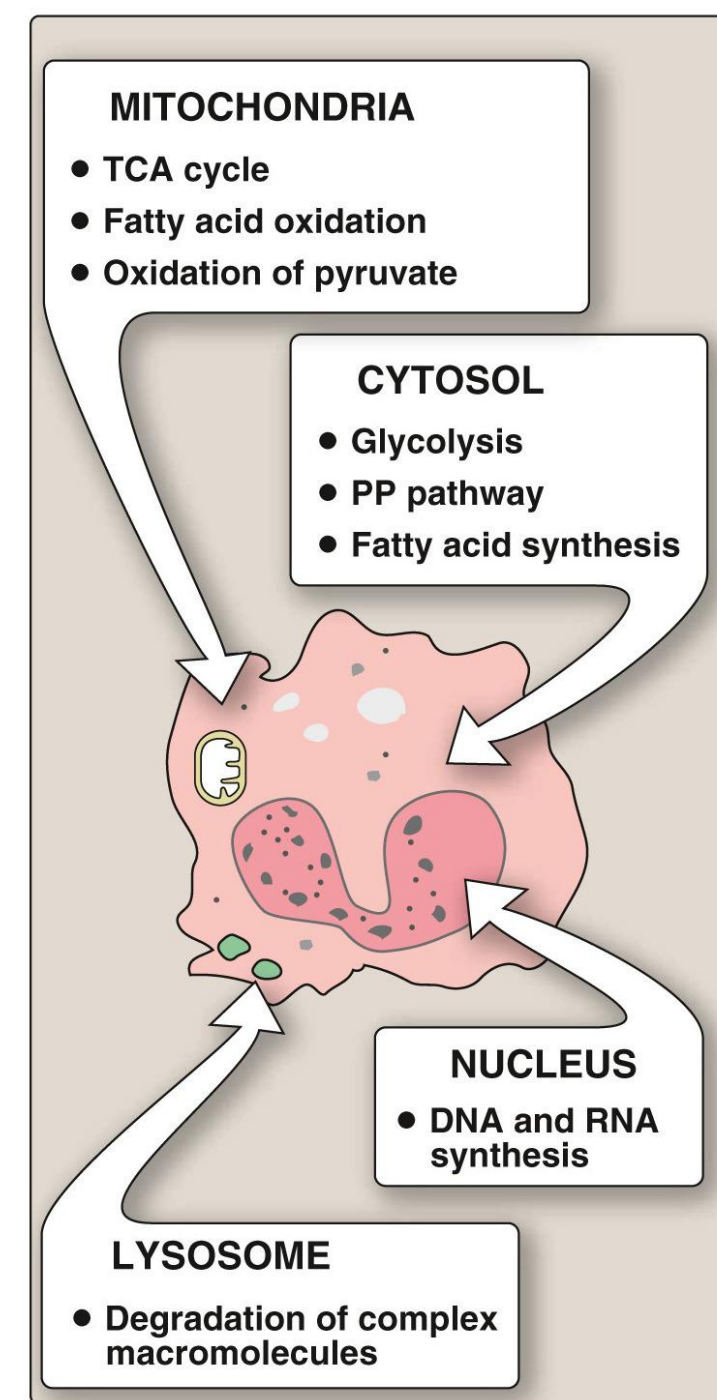


Purpose of pathway:
Regulation: Fed vs fasting
Disorders

Metabolic pathways

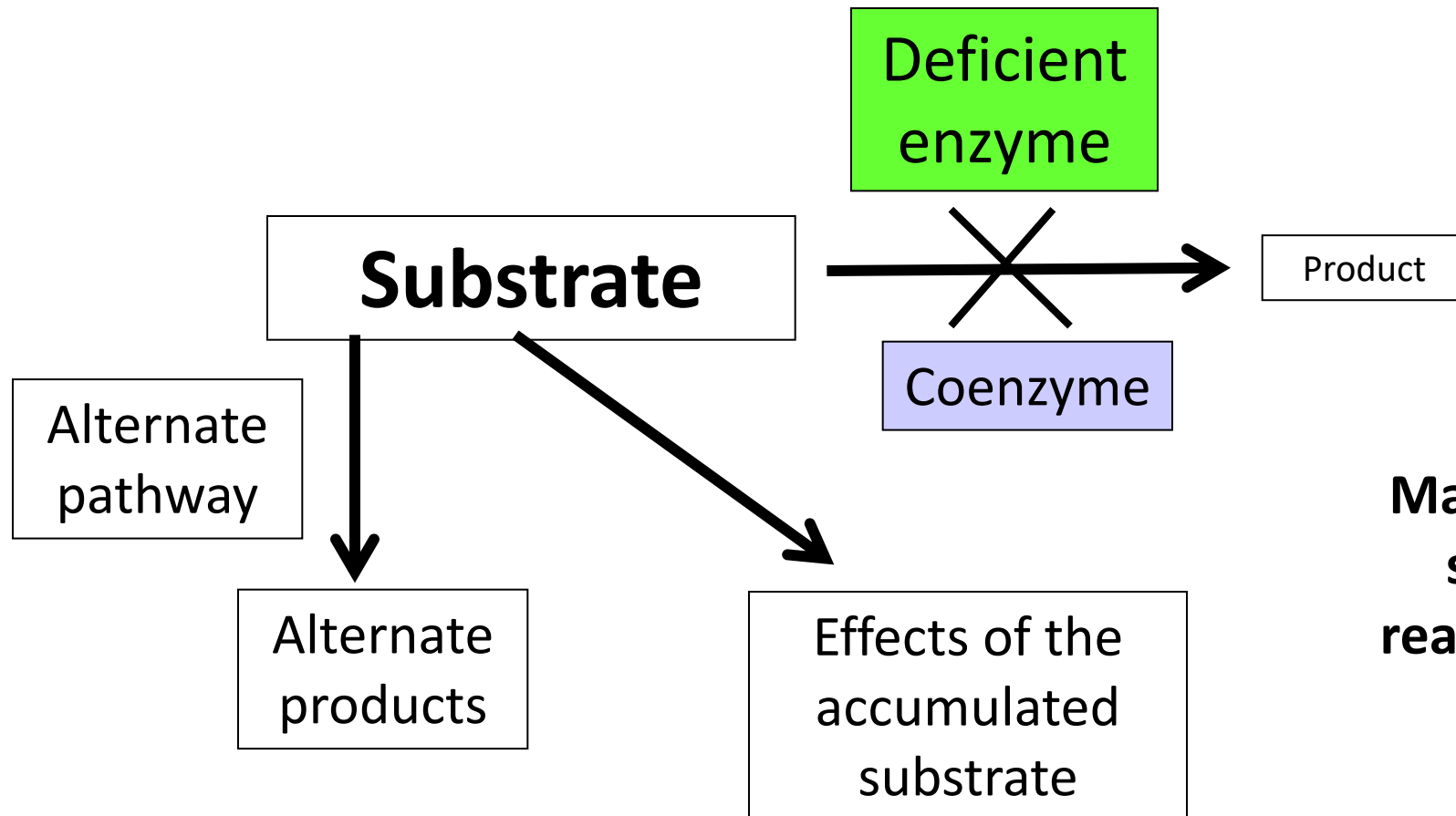


Tissues in which pathway is active and **intracellular location**



Overview of effects of inherited enzyme deficiencies

Understand and analyze the clinical effects of the enzyme deficiency and the laboratory tests for each of the disorder



Make a table for water soluble vitamins – reactions requiring that vitamin

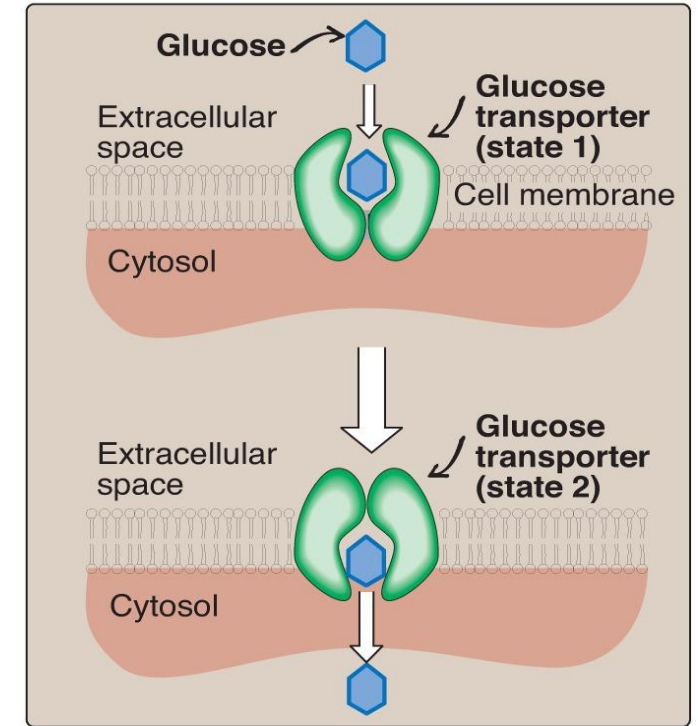
Vitamin	Other Names	Active Form	Function
Vitamin B ₉	Folic acid	Tetrahydro-folic acid	Transfer one-carbon units; synthesis of methionine, serine, purine nucleotides, and thymidine monophosphate
Vitamin B ₁₂	Cobalamin	Methylcobalamin Deoxyadenosyl cobalamin	Coenzyme for reactions: Homocysteine → methionine Methylmalonyl CoA → succinyl CoA
Vitamin C	Ascorbic acid	Ascorbic acid	Antioxidant Coenzyme for hydroxylation reactions, for example: In procollagen: Proline → hydroxyproline Lysine → hydroxylysine
Vitamin B ₆	Pyridoxine Pyridoxamine Pyridoxal	Pyridoxal phosphate	Coenzyme for enzymes, particularly in amino acid metabolism
Vitamin B ₁	Thiamine	Thiamine pyrophosphate	Coenzyme of enzymes catalyzing: Pyruvate → acetyl CoA α-Ketoglutarate → Succinyl CoA Ribose 5-P + xylulose 5-P → Sedoheptulose 7-P + Glyceraldehyde 3-P Branched-chain α-keto acid oxidation
Vitamin B ₃	Niacin Nicotinic acid	NAD ⁺ , NADP ⁺	Electron transfer
Vitamin B ₂	Riboflavin	FMN, FAD	Electron transfer
Vitamin B ₇	Biotin	Enzyme-bound biotin	Carboxylation reactions
Vitamin B ₅	Pantothenic acid	Coenzyme A	Acyl carrier
WATER SOLUBLE			
Vitamin A	Retinol Retinal Retinoic acid β-Carotene	Retinol Retinal Retinoic acid	Maintenance of reproduction Vision Promotion of growth Differentiation and maintenance of epithelial tissues Gene expression
Vitamin D	Cholecalciferol Ergocalciferol	1,25-Dihydroxy-cholecalciferol	Calcium uptake Gene expression
Vitamin K	Menadione Menaquinone Phylloquinone	Menadione Menaquinone Phylloquinone	γ-Carboxylation of glutamate residues in clotting and other proteins
Vitamin E	α-Tocopherol	Any of several tocopherol derivatives	Antioxidant

Lippincott’s Biochemistry Fig 28.29

Deficiency	Signs and Symptoms	Toxicity	Notes
Megaloblastic anemia Neural tube defects	Anemia Birth defects	None	Administration of high levels of folate can mask vitamin B ₁₂ deficiency
Pernicious anemia Dementia Spinal degeneration	Megaloblastic anemia Neuropsychiatric symptoms	None	Pernicious anemia is treated with intramuscular or high-dose oral vitamin B ₁₂
Scurvy	Sore, spongy gums Loose teeth Poor wound healing Bleeding	None	Benefits of supplementation not established in controlled trials
Rare	Glossitis Neuropathy	Yes	Deficiency can be induced by isoniazid Sensory neuropathy occurs at high doses
Beriberi Wernicke-Korsakoff syndrome (most common in alcoholism)	Peripheral neuropathy (dry form), edema and cardiomyopathy (wet form) Confusion, ataxia, memory loss, hallucinations, dysregulated eye movements	None	—
Pellagra	Dermatitis Diarrhea Dementia	None	High doses of niacin used to treat hyperlipidemia
Rare	Dermatitis Angular stomatitis	None	—
Rare	Dermatitis	None	Consumption of large amounts of raw egg whites (which contains a protein, avidin, that binds biotin) can induce a biotin deficiency
Rare	—	None	—
WATER SOLUBLE			
Night blindness Xerophthalmia Infertility Growth retardation	Increased visual threshold Dryness of cornea	Yes	β-Carotene not acutely toxic, but supplementation is not recommended Excess vitamin A can increase incidence of fractures
Rickets (in children) Osteomalacia (in adults)	Soft, pliable bones	Yes	Vitamin D is not a true vitamin because it can be synthesized in skin; application of sunscreen lotions or presence of dark skin color decreases this synthesis.
Newborn Rare in adults	Bleeding	Rare	Vitamin K produced by intestinal bacteria. Vitamin K deficiency common in newborns Intramuscular treatment with vitamin K is recommended at birth
Rare	Red blood cell fragility leads to hemolytic anemia	None	Benefits of supplementation for disease prevention not established in controlled trials

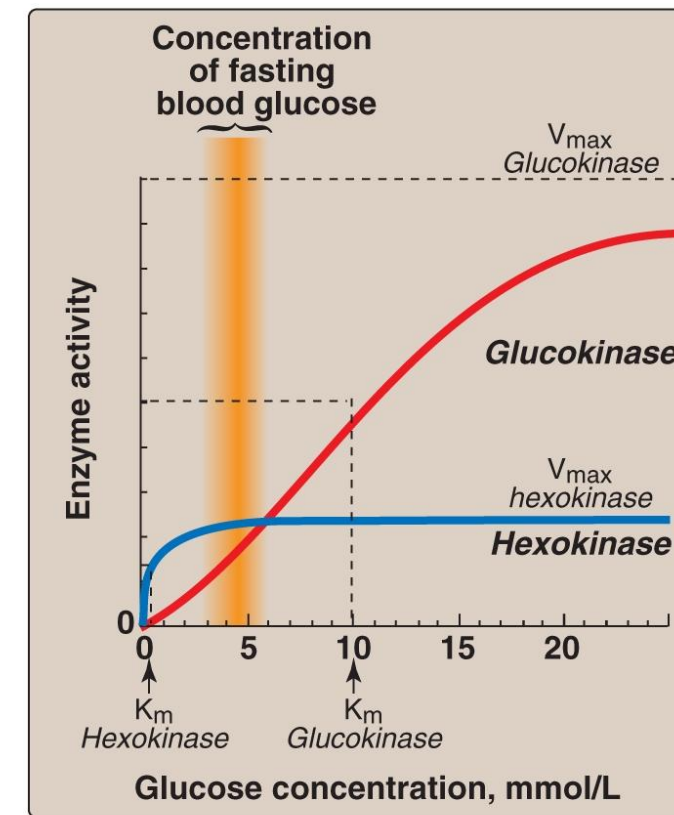
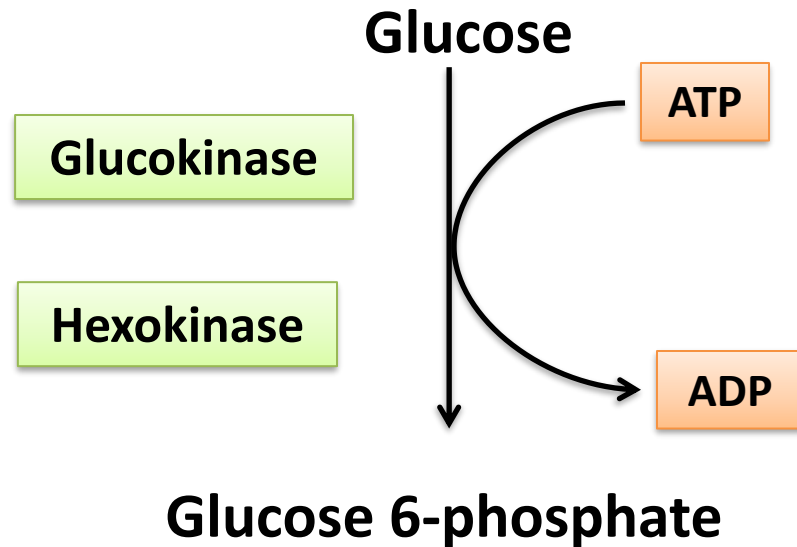
Fate of absorbed glucose: Glucose transport into cells

- GLUT (facilitated diffusion)
 - **GLUT 2** – liver
 - **GLUT 1 and 3** – neurons and brain
 - GLUT 1 - erythrocytes
 - **GLUT 4** - adipose tissue and muscle (Insulin-responsive)



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Formation of glucose 6 phosphate

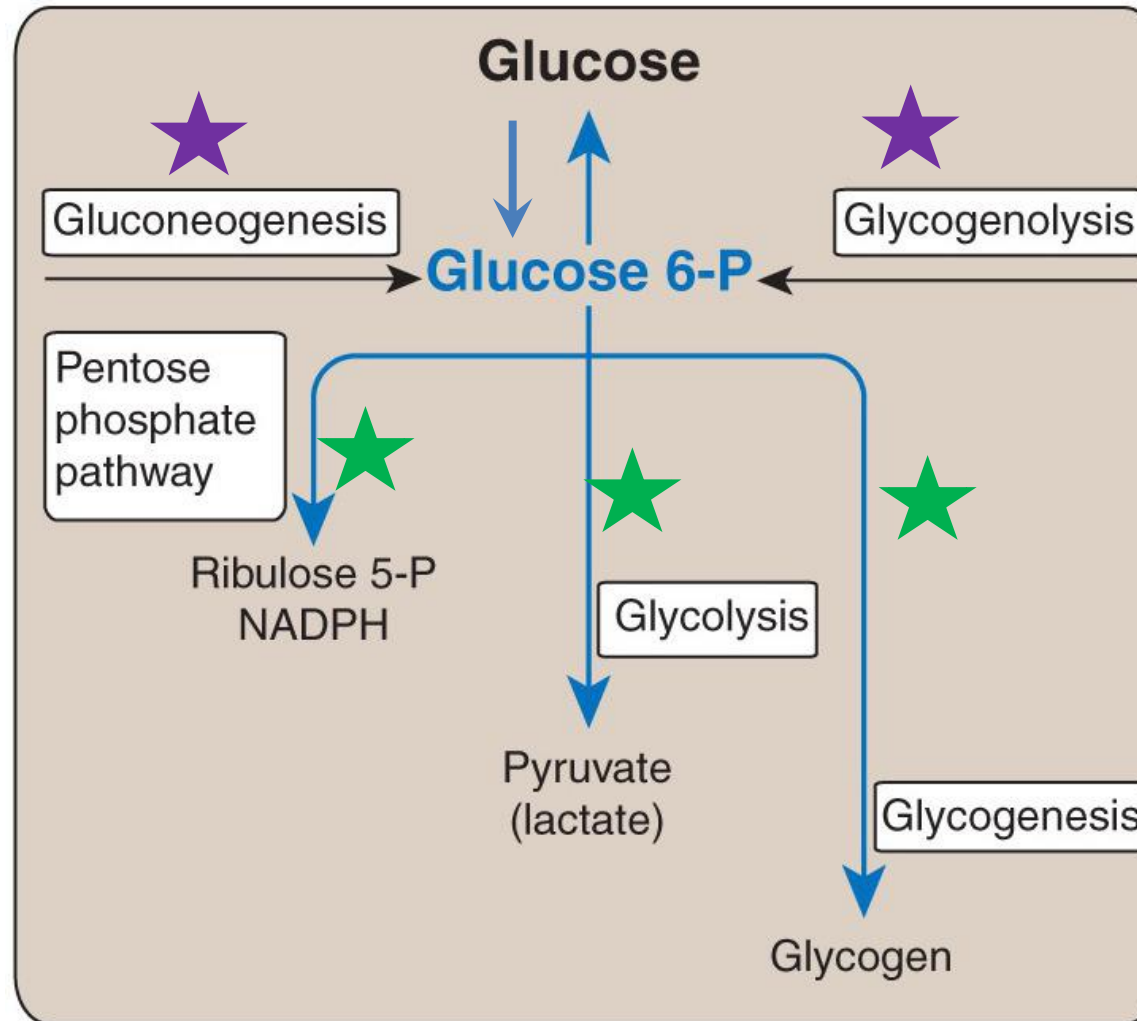


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- Glucokinase: **liver and pancreatic β -cells of pancreas (high K_m)**
 - More active when blood glucose is elevated; 'Glucose sensor'
- Other tissues – hexokinase (low K_m)
- Glucokinase mutations: Hyperglycemia and Monogenic diabetes (previously MODY-2)
 - β -cells of pancreas cannot respond to high blood glucose levels!!!

Fates of glucose 6-phosphate in liver

Non-carbohydrate
sources (Lactate,
amino acids, glycerol)



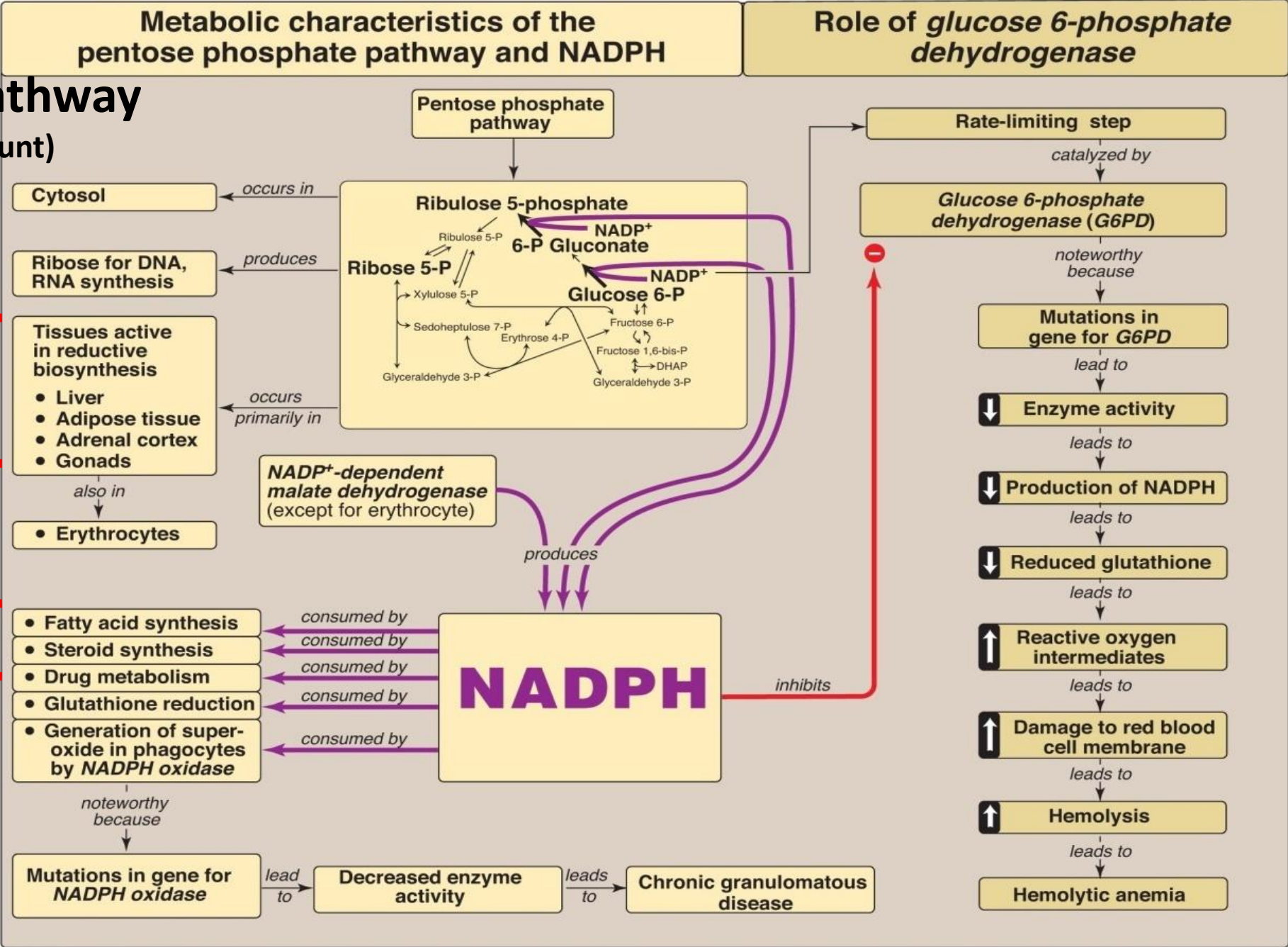
Glycogen

Pathways active when

- Insulin is high (fed state) ★
- Glucagon is high (fasting) ★

Pentose phosphate pathway

(Hexose monophosphate shunt)



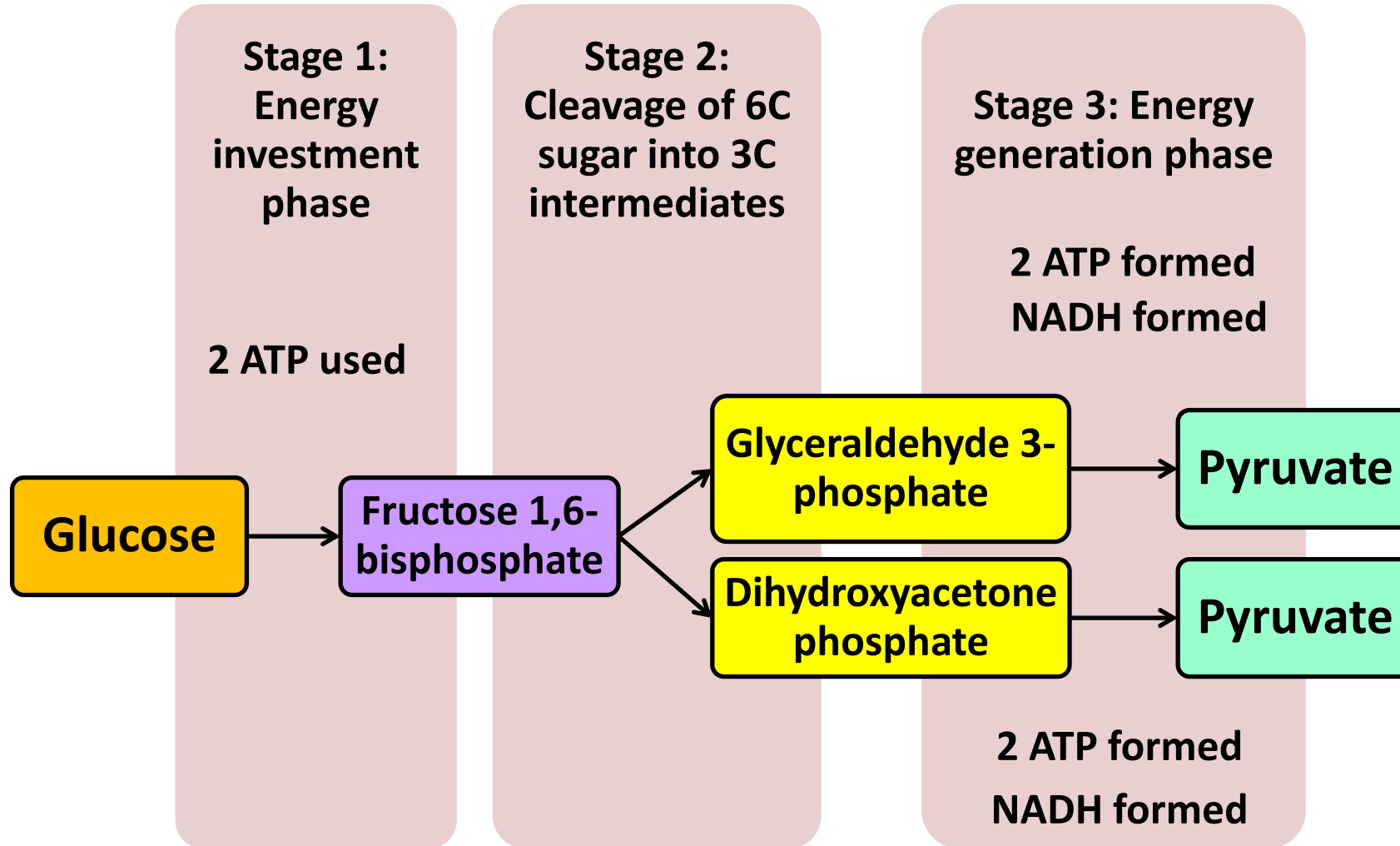
Pentose phosphate pathway: Points to review

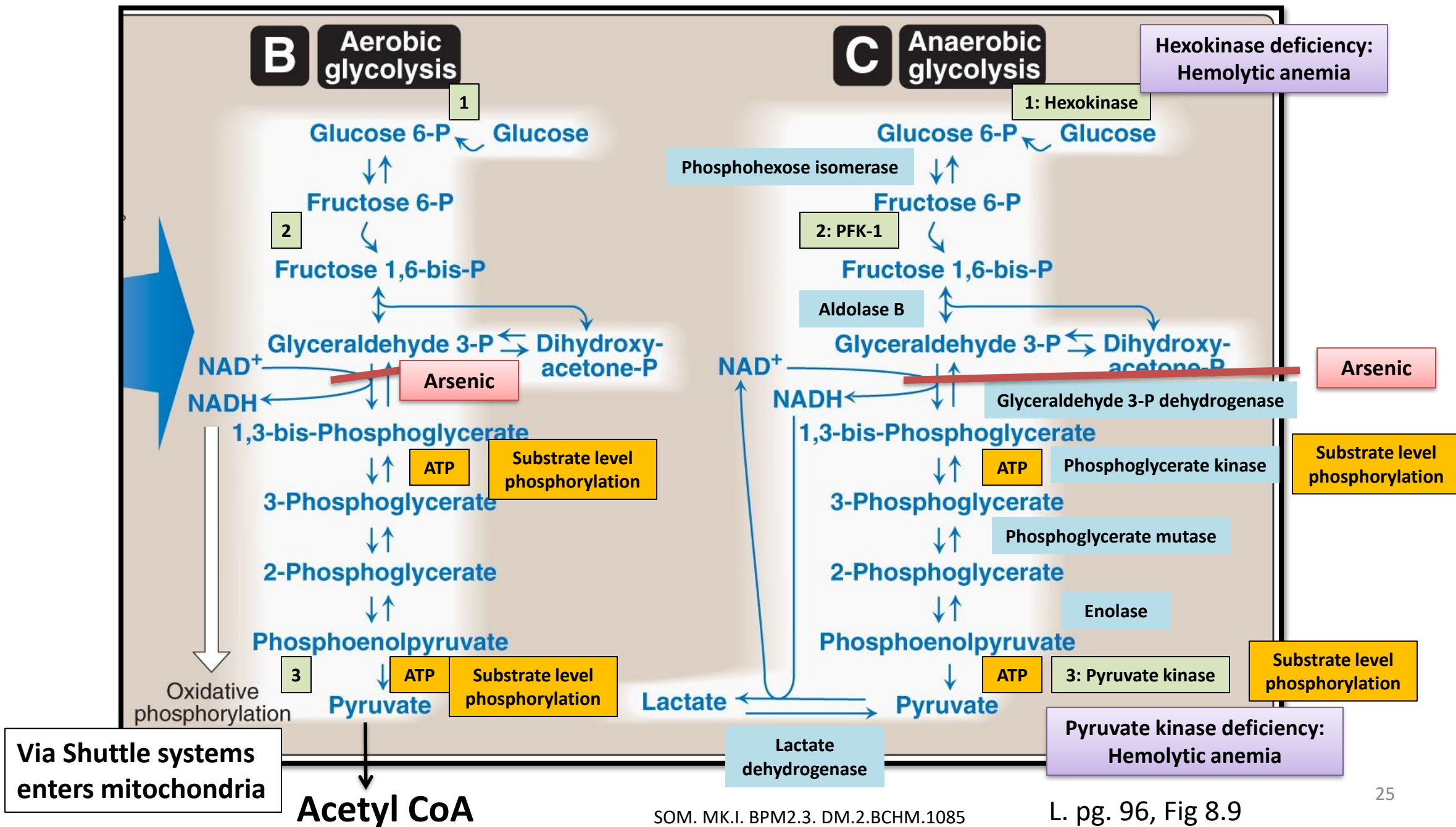
- **CYTOSOL**
- **Glucose 6-phosphate dehydrogenase** (key enzyme)
 - **Does NOT form ATP (energy)**
 - **G6PD deficiency** → Hemolytic anemia (due to low NADPH)
- Forms **NADPH**
 - **Fatty acid biosynthesis, cholesterol biosynthesis, and steroid hormone synthesis** in liver, adipocytes, and endocrine glands
 - Malic enzyme also forms NADPH in these tissues
- Forms Ribose phosphate – **purine and pyrimidine synthesis**
- **Transketolase** requires **TPP (vitamin B1)**
 - Thiamine deficiency: Low transketolase activity

Glycolysis

- Generates **ATP** (energy)
 - In brain (fed and fasted state), liver (fed state)
 - Tissues that lack mitochondria or hypoxic (Only pathway for energy generation)
 - Actively contracting skeletal muscle (Glucose obtained from glycogen or plasma glucose)
- Generate **ATP (energy)** in presence or absence of O_2 (aerobic and anaerobic) and in presence or absence of mitochondria
- Glycolytic enzymes are present in **cytosol**

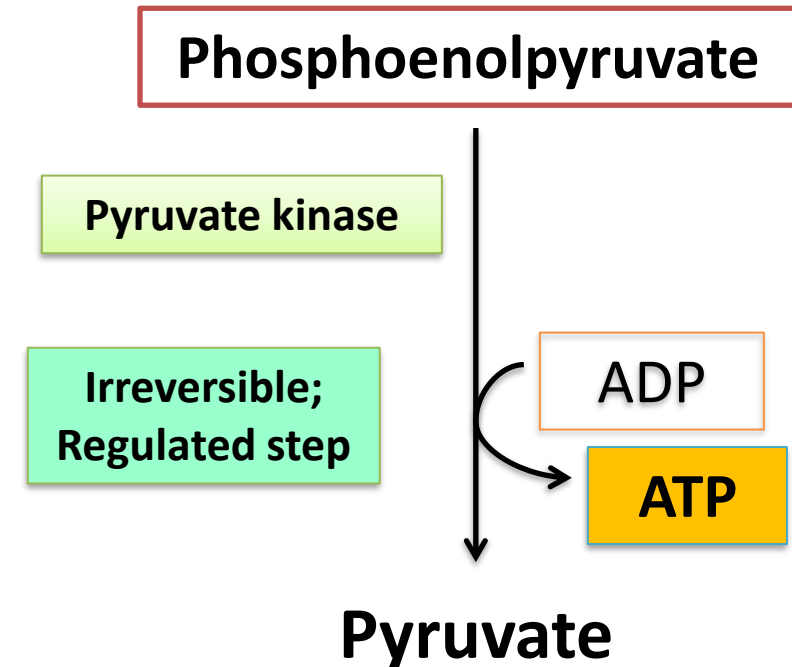
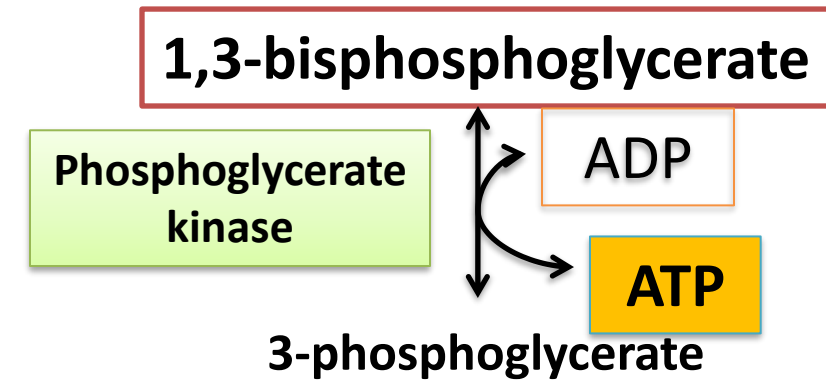
Overview of phases of glycolysis



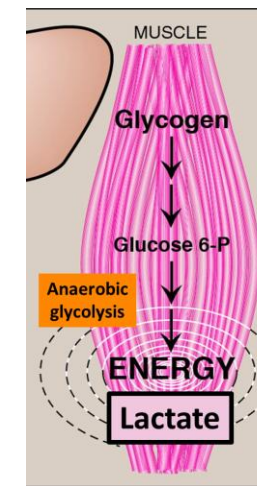
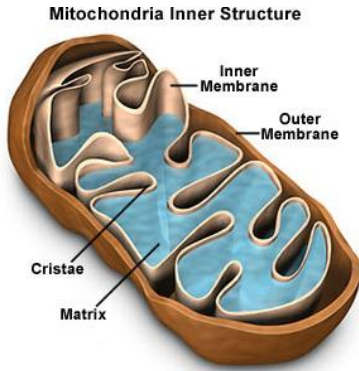


ATP formed: Substrate level phosphorylation

- Directly form ATP from ADP
- DO NOT require Electron Transport chain or mitochondria or Oxygen
- ONLY reactions that form ATP in cells that lack mitochondria and in hypoxic conditions
- High energy compounds
 - 1,3-Bisphosphoglycerate
 - Phosphoenolpyruvate



Fates of pyruvate



Pyruvate

Aerobic (oxygen and mitochondria)

Anaerobic (no mitochondria/ no oxygen)

Acetyl CoA (Pyruvate dehydrogenase complex – requires TPP)

Lactate (Lactate dehydrogenase)

Cytosolic enzyme

Enters TCA cycle

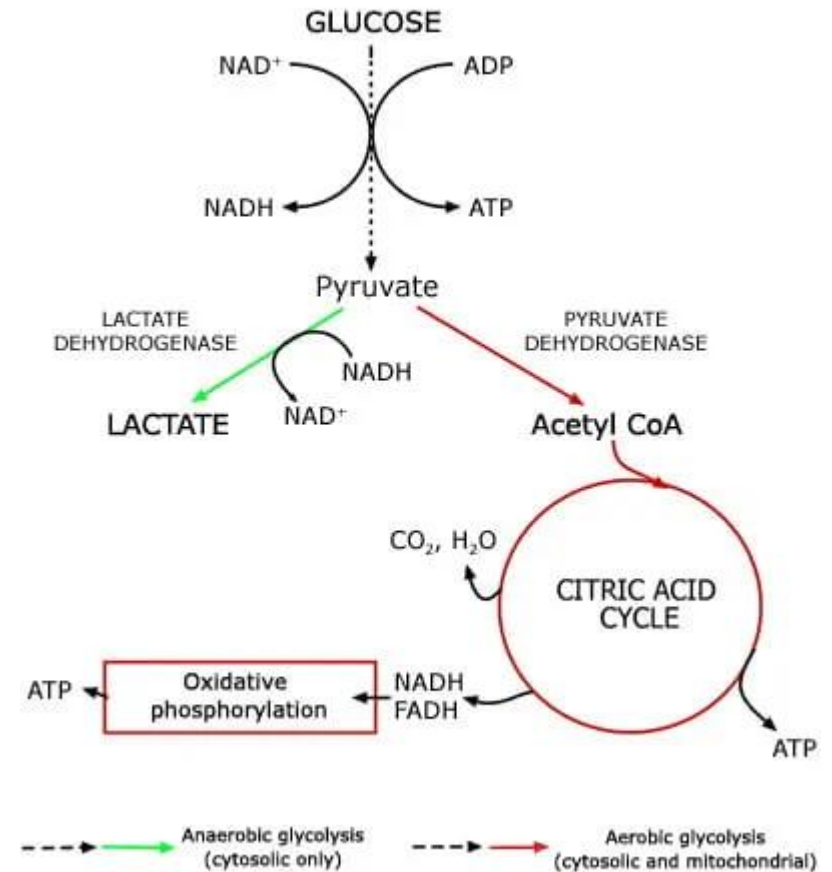
Goes to liver via Cori cycle

Increased lactate production:
Absence of mitochondria (red blood cells, lens, cornea)
Poorly vascularized tissues or hypoxia (contracting skeletal muscle, cardiac muscle hypoxia)
Pyruvate dehydrogenase deficiency or thiamine deficiency

Mitochondrial enzyme

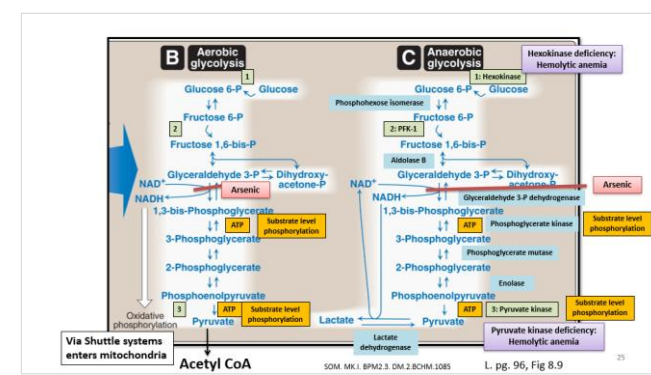
Fates of pyruvate

- **Increased lactate production:**
 - **Absence of mitochondria** (red blood cells, lens, cornea)
 - **Poorly vascularized tissues or hypoxia** (contracting skeletal muscle, cardiac muscle hypoxia)
 - **Pyruvate dehydrogenase deficiency or thiamine deficiency**

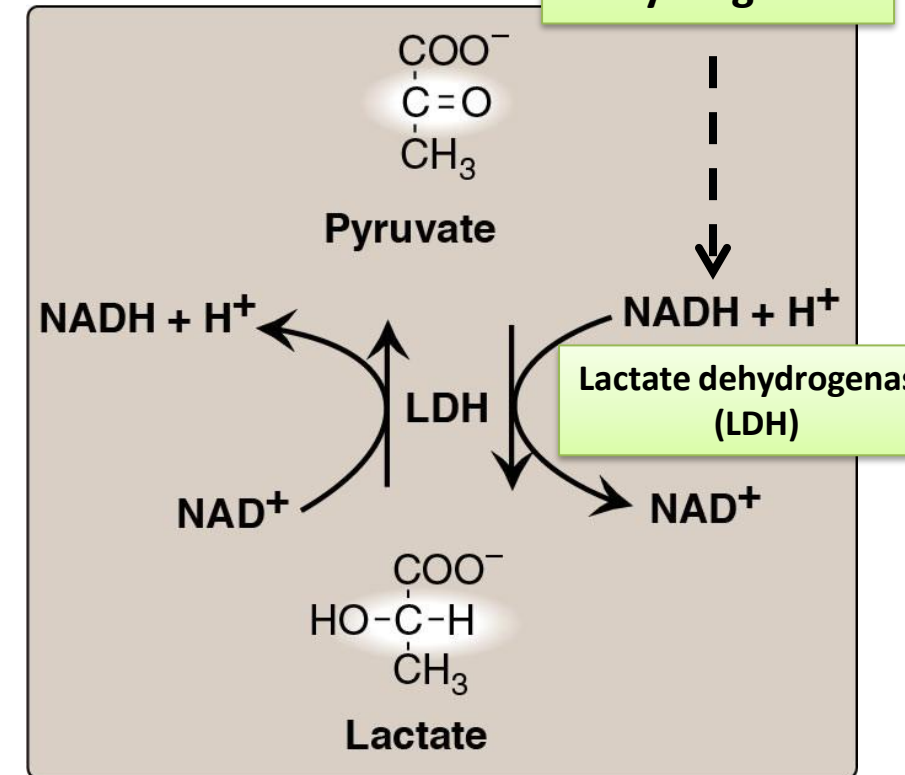


Anaerobic fate of pyruvate

- NADH formed by glyceraldehyde 3-phosphate dehydrogenase is reoxidized to form NAD^+
 - Allows glycolysis to proceed in absence of mitochondria
- LDH is cytosolic enzyme
- Direction of reaction: **NADH/NAD^+ ratio**
 - Higher NADH, increases lactate
- Identify states of high NADH/NAD^+ ratio
 - Anaerobic
 - Binge alcohol intake
- Contrast between muscle (fast-twitch) and liver



Glyceraldehyde 3-phosphate dehydrogenase



Energetics under aerobic conditions

(Oxygen abundant/ mitochondria abundant)

- Overall reaction: Glucose → 2 Pyruvate
- 2 NADH + H⁺ (glyceraldehyde 3-phosphate dehydrogenase) = 2 × 3 = 6 ATP
- (1 NADH = 2.5 ATP and 1 FADH₂ = 1.5 ATP; Approximate 1 NADH = 3 ATP and 1 FADH₂ = 2 ATP)
- NADH in **cytosol** by glycolysis transported into mitochondria by **shuttle systems** (**malate-aspartate** or **glycerol-phosphate shuttle**) – Refer Shuttles and OxPhos
- 4 ATP (substrate level phosphorylation)
 - (2 ATP from each glyceraldehyde 3-phosphate)
- -2 ATP used
- Total = 8 ATP

Energetics under anaerobic conditions (Lack of oxygen/ no mitochondria)

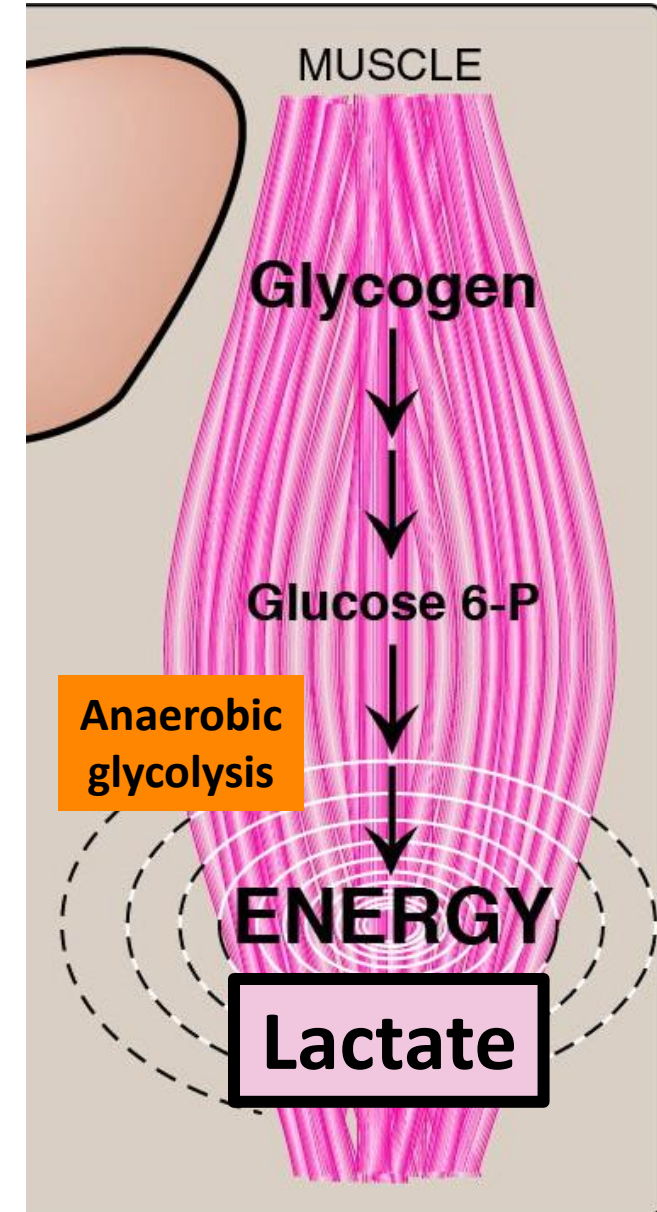
- Overall reaction Glucose → 2 Lactate
- 2 NADH + H⁺ (glyceraldehyde 3-phosphate dehydrogenase) used in lactate dehydrogenase
- 4 ATP (**substrate level phosphorylation**) (2 ATP from a glyceraldehyde 3-phosphate)
- -2 ATP used
- Total = 2 ATP

Lactate formation during anaerobic exercise in muscle

- **Lactate threshold and endurance training:**

Weight loss advice

- 60-70% of maximum heart rate: Fat oxidation by muscle
 - 80-85% of MHR: Glycolysis under anaerobic conditions forms lactate
- Anaerobic (strenuous) activity, blood **lactate levels increase**
 - Lactate in muscle → pH drops and cramps



Aerobic and anaerobic exercise: Fuel sources

Aerobic/endurance (cardio) exercise



- Oxygen present; Abundant mitochondria
- Glycogenolysis is active
- Glycolysis is active
- Krebs' cycle and electron transport chain are both active, producing ample ATP
- Fatty acid oxidation
- ATP production is maximal

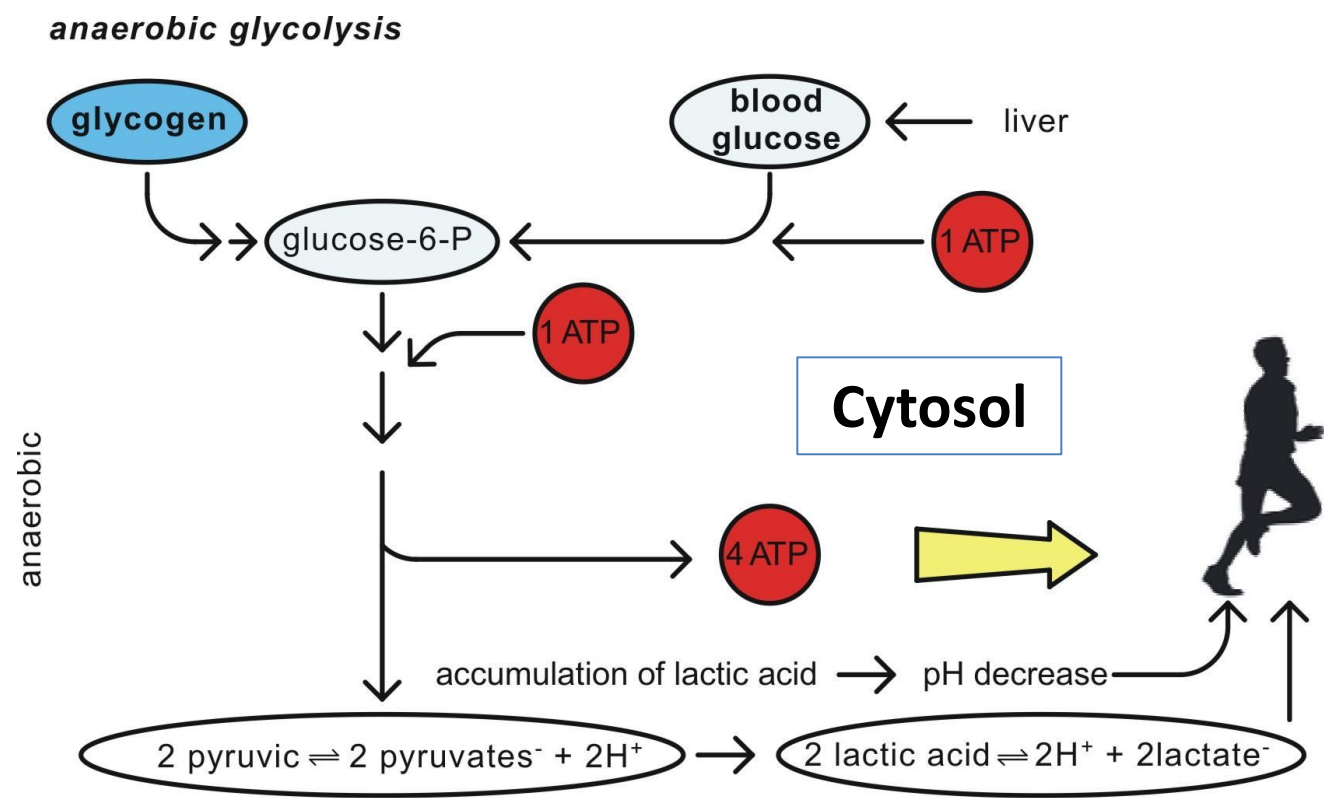
Anaerobic/ strenuous/ ischemic exercise



- Oxygen absent; Cytosol
- Glycogenolysis is active, but of short duration
- Glycolysis is active
- Pyruvate reduced to LACTATE
- ATP production is rapid but minimal

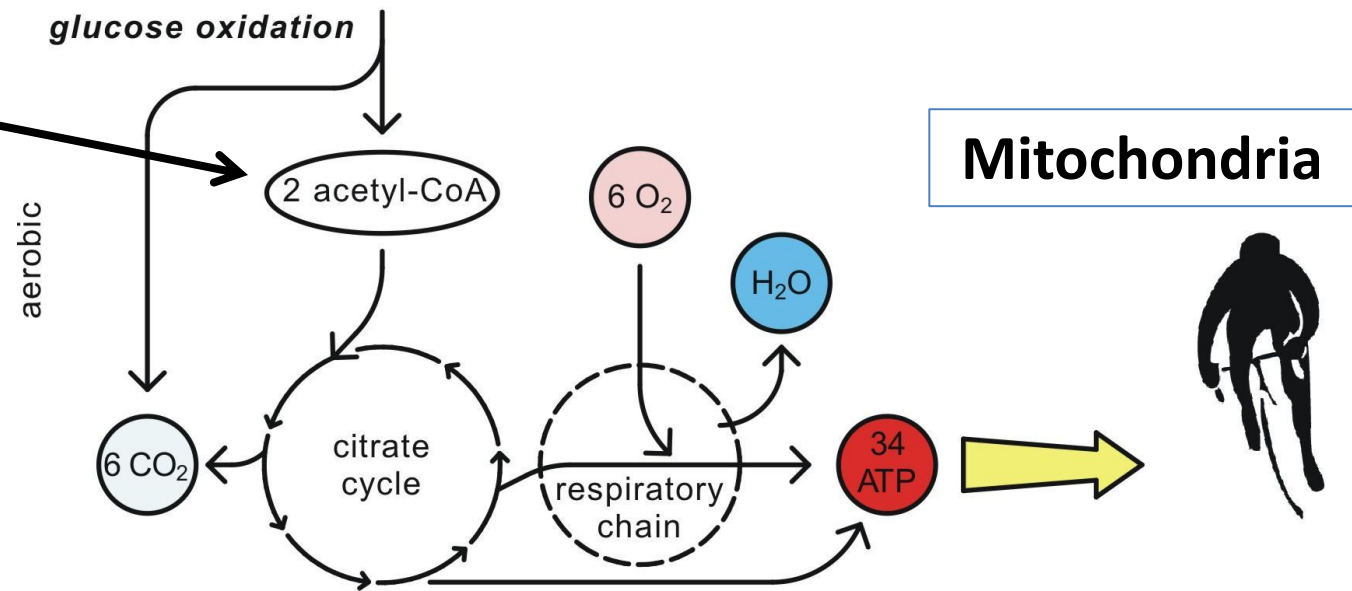
Metabolism in muscle: Aerobic and anaerobic

Glycogenolysis and Glycolysis (Anaerobic)



Fatty acid oxidation

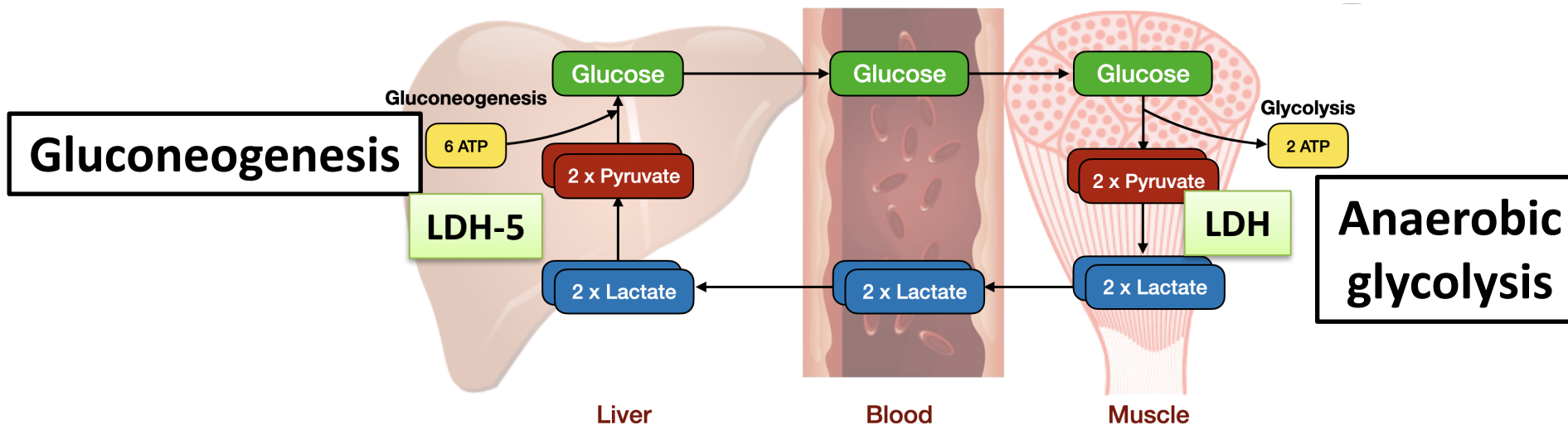
Aerobic: Fatty acid oxidation and Krebs cycle



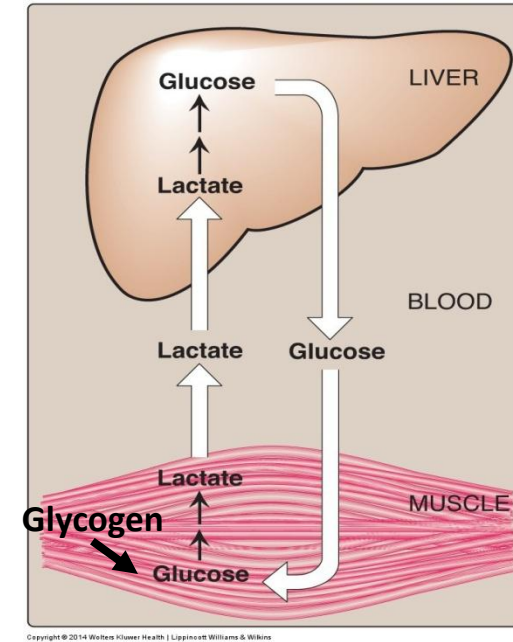
Fate of lactate: Cori cycle

Glucose (liver) and lactate (Skeletal muscle; RBC)

Lactate from skeletal muscle and RBC, transported to liver for gluconeogenesis

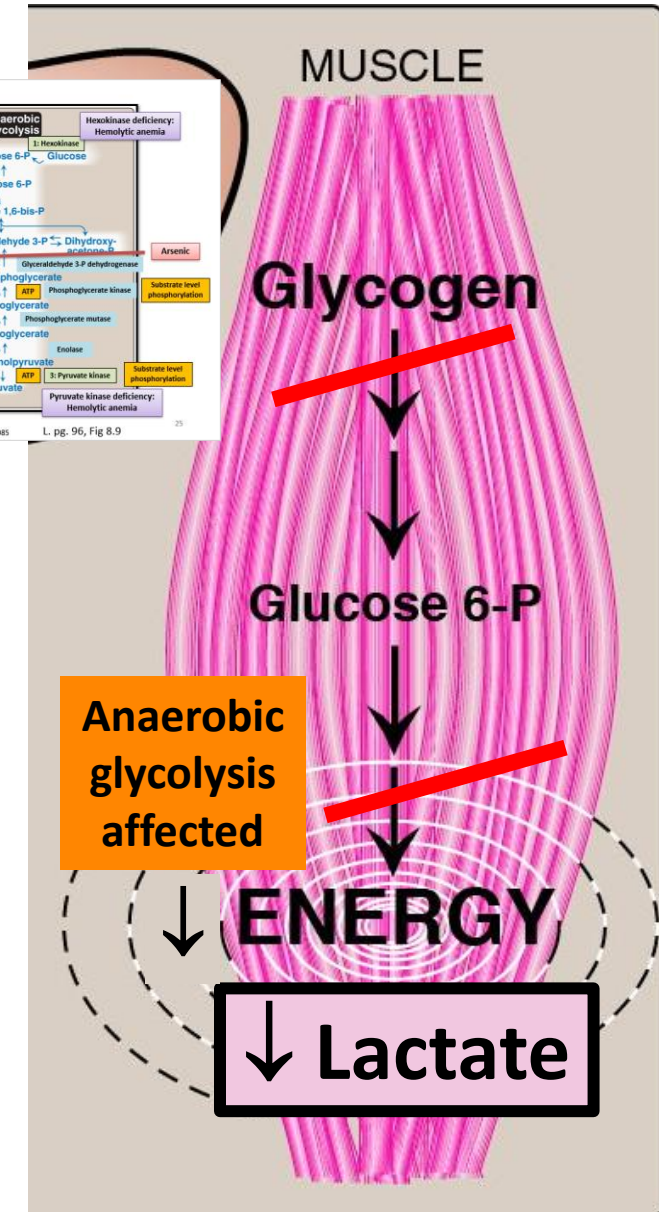
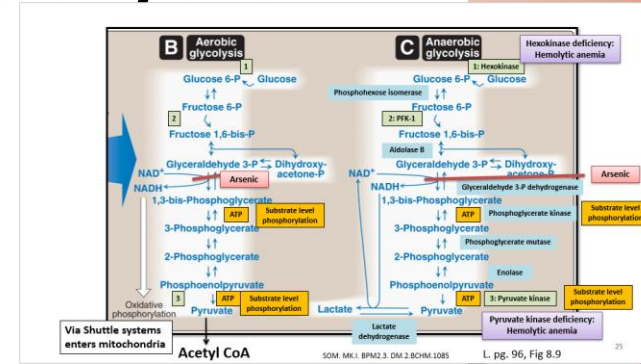


**Red blood cell,
Skeletal muscle**



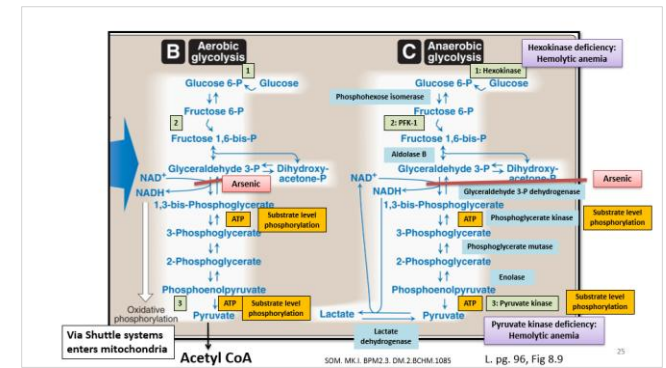
Defects in muscle glycolysis

- Muscle **PFK-1 deficiency** (Tarui disease; Glycogen storage disease Type VII)
- **Muscle cramps** (due to low ATP) during high-intensity anaerobic exercise; hemolysis due to RBC glycolytic defect
- Myoglobinuria and high serum CK-MM levels following high-intensity exercise
- **NO increase in blood lactate levels** with high-intensity anaerobic exercise (forearm exercise test)
- Aerobic exercise tolerated (Oxidation of fatty acids not affected)

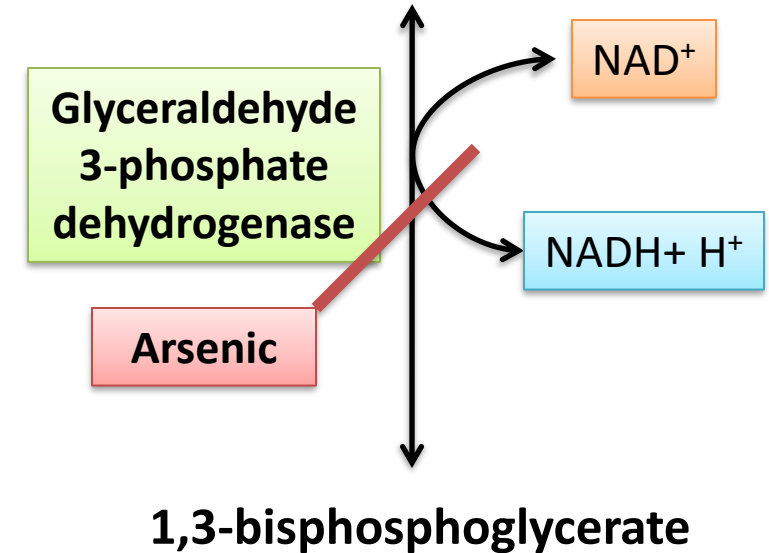


Arsenic poisoning

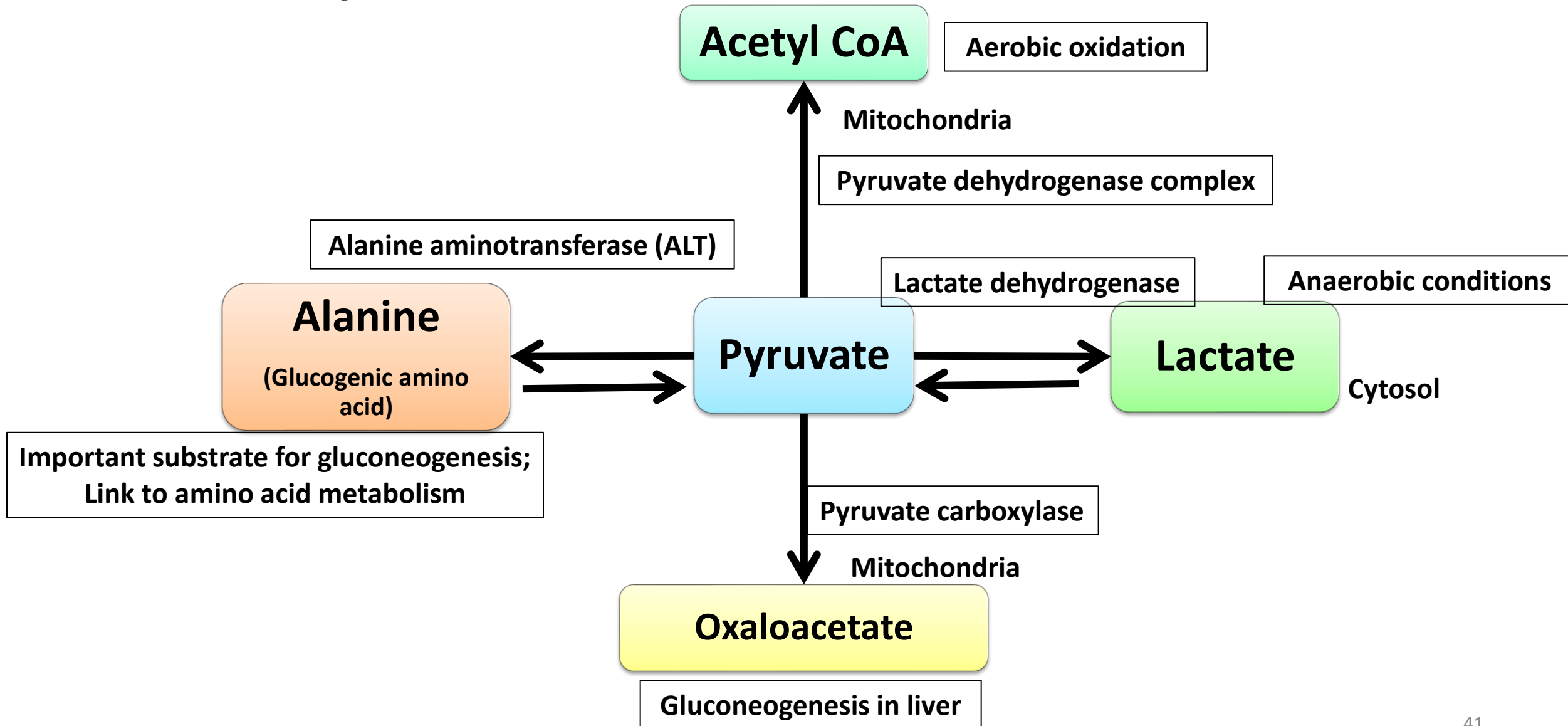
- **Glyceraldehyde 3-phosphate dehydrogenase**
- PDH complex and α -ketoglutarate dehydrogenase complex (mitochondrial enzymes)
- Fo-F1 ATP synthase also affected



Glyceraldehyde 3-phosphate



Fates of Pyruvate

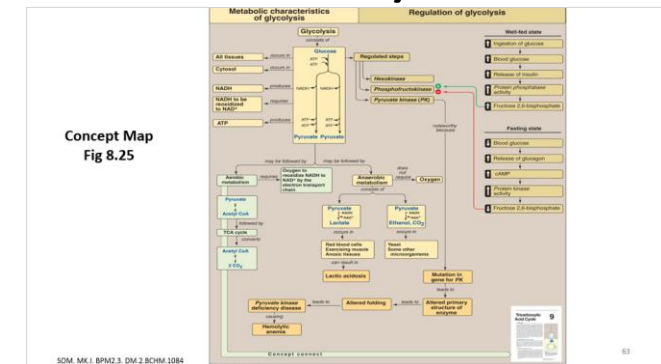


Lactic acidosis

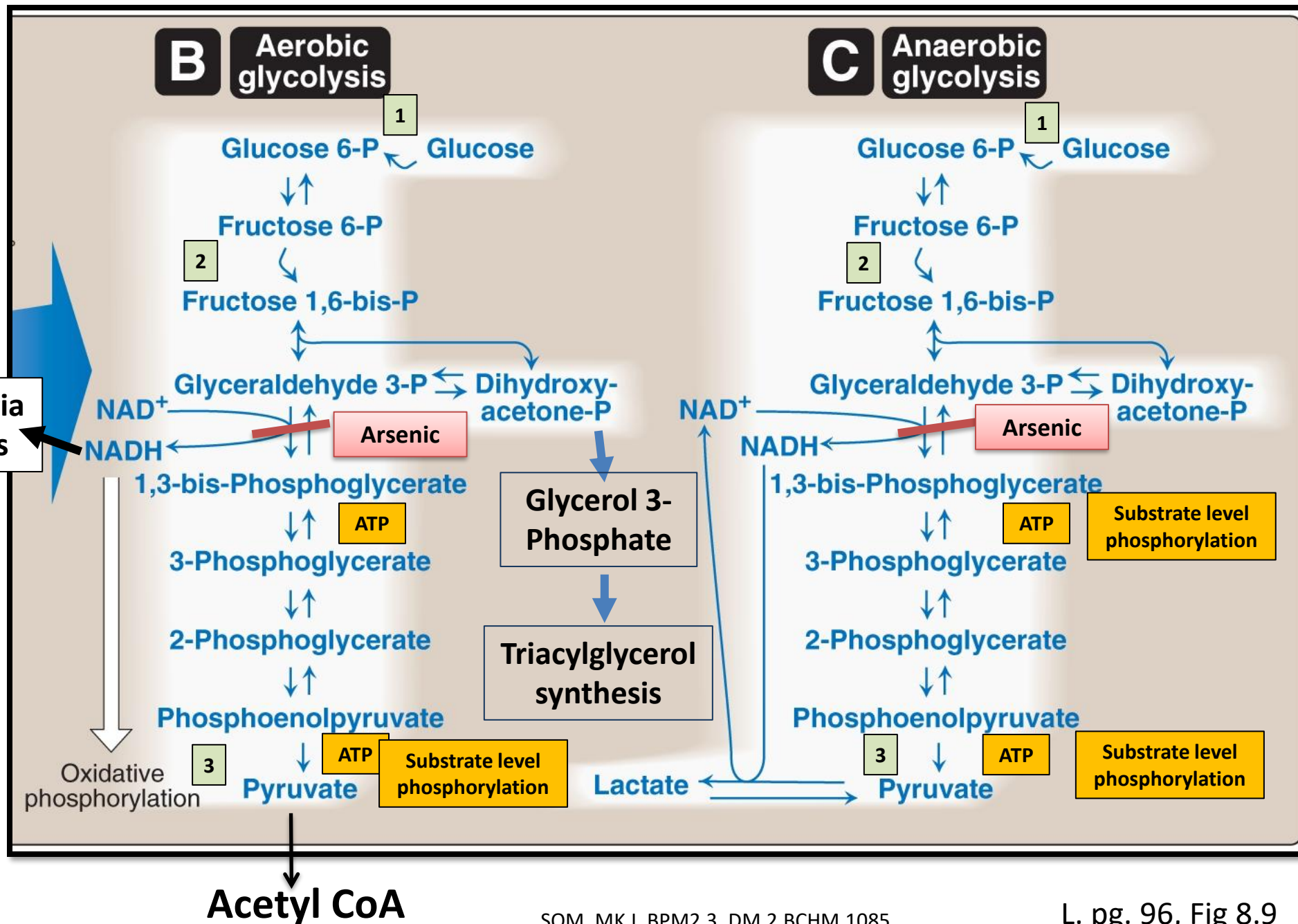
- **High anion gap metabolic acidosis**
 - pH=low; HCO_3^- decreased; PCO_2 decreased (compensation)
 - High anion gap; **Elevated lactate levels** (unmeasured anion)
- Conditions of Lactic acidosis
 - Increased NADH/NAD⁺ ratio: **Binge alcohol consumption**
 - **Pyruvate dehydrogenase** deficiency (Leigh disease)
 - **Thiamine deficiency** (pyruvate dehydrogenase complex)
 - **Gluconeogenesis defect** – von Gierke disease – Defect in Cori cycle
 - **Decreased blood supply** (circulatory shock): hypoxia and anaerobic metabolism
 - **Pulmonary embolism** (poor oxygenation)

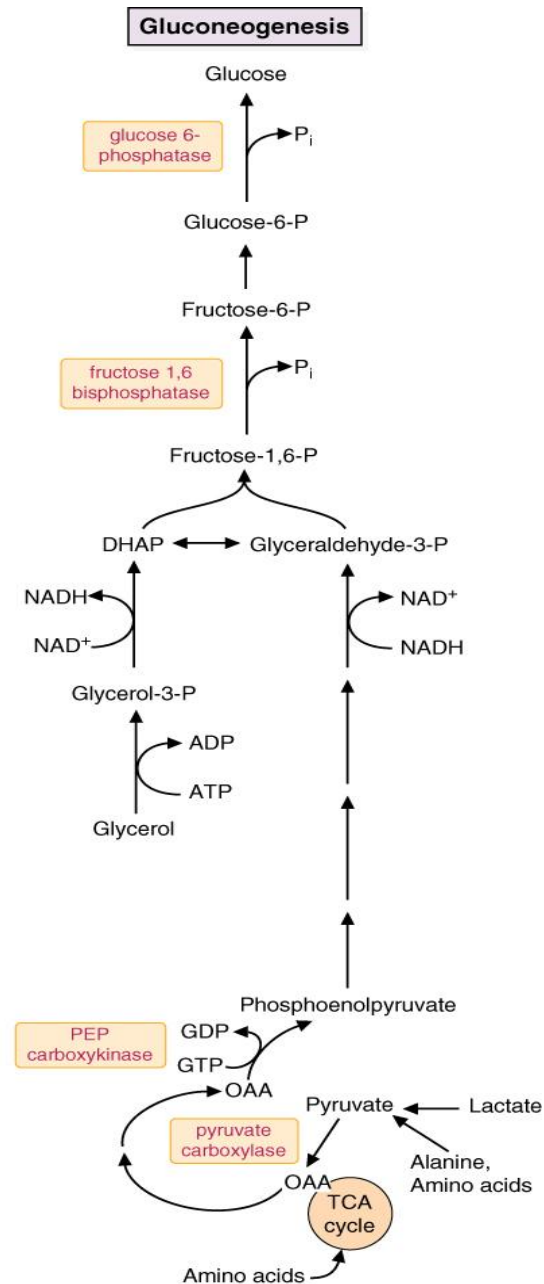
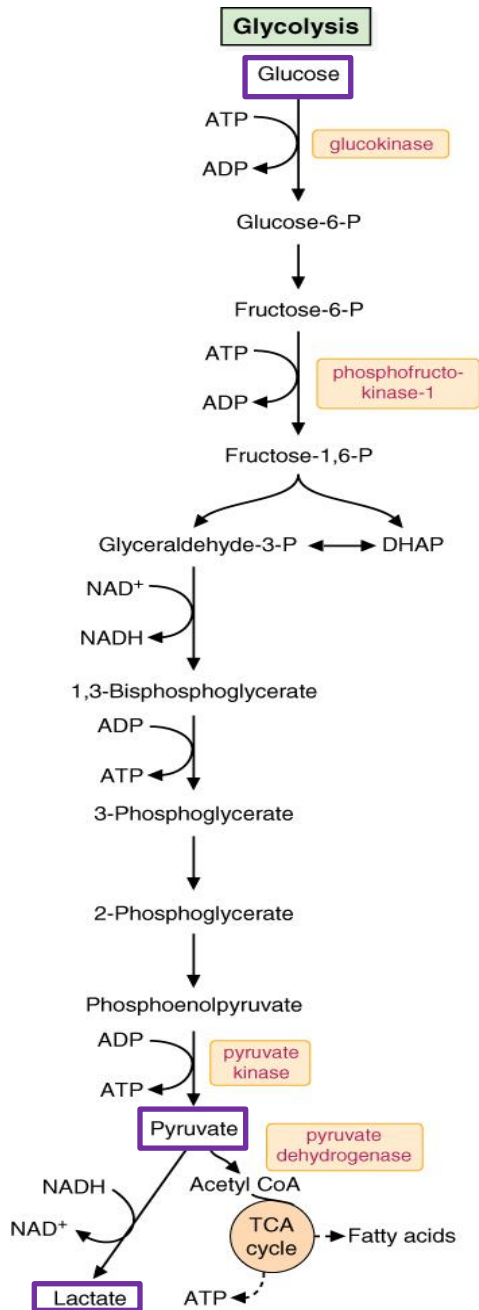
Significance of glycolysis

- Aerobic Tissues (brain, skeletal muscle) Glucose → pyruvate → acetyl CoA → TCA cycle
- **Adipose tissue: Dihydroxyacetone phosphate** (forms glycerol 3-P) used for **triacylglycerol formation** (Link to lipid metabolism)
- Tissues of eye: cornea, lens (no mitochondria)
- Tumor cells
- **Wound and healing injured tissue**
- Red blood cells, anaerobic glycolysis
 - Only source of energy (ATP)
 - Forms 2,3 bisphosphoglycerate (2,3 BPG)



Enters mitochondria
via Shuttle systems





Gluconeogenesis vs Glycolysis

- Gluconeogenesis almost reverse of glycolysis: except for three irreversible glycolysis reactions
- **Four irreversible reactions in gluconeogenesis**
- All other steps are reversible
- Only one pathway can be active while the other is inhibited (under a specific metabolic condition) → **Reciprocal regulation**

Regulation of glycolysis in liver

- Liver glycolysis and gluconeogenesis are **reciprocally** regulated
- Glycolysis active when high blood glucose levels/ high insulin
- Gluconeogenesis: Active at low blood glucose levels/ high glucagon
- Irreversible/ regulated reactions:

– Glucokinase

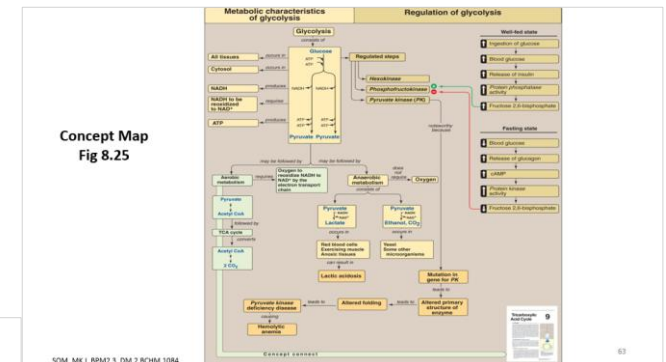
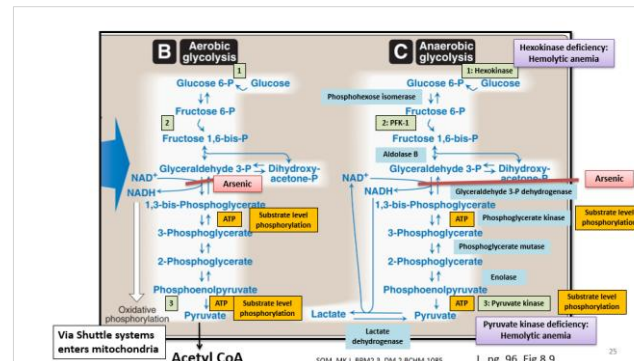
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– Phosphofructokinase-1 (PFK-1)

2

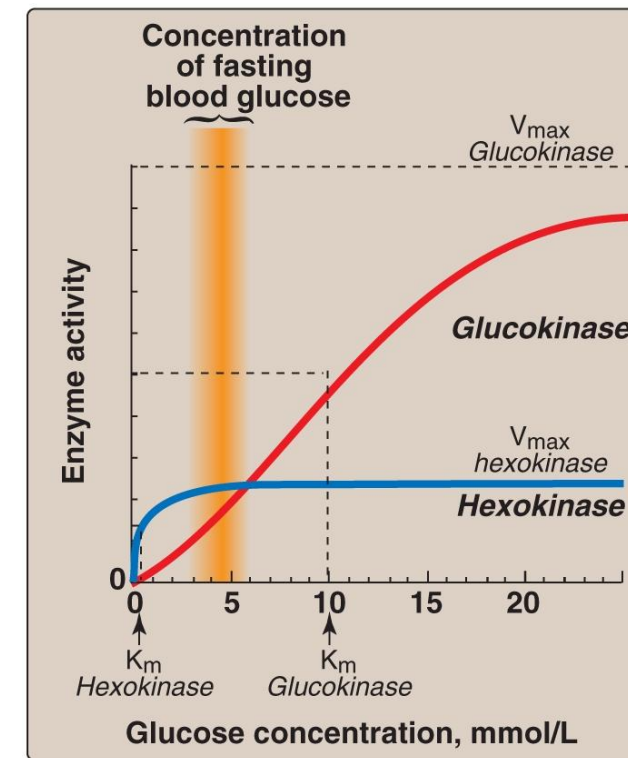
– Pyruvate kinase

3

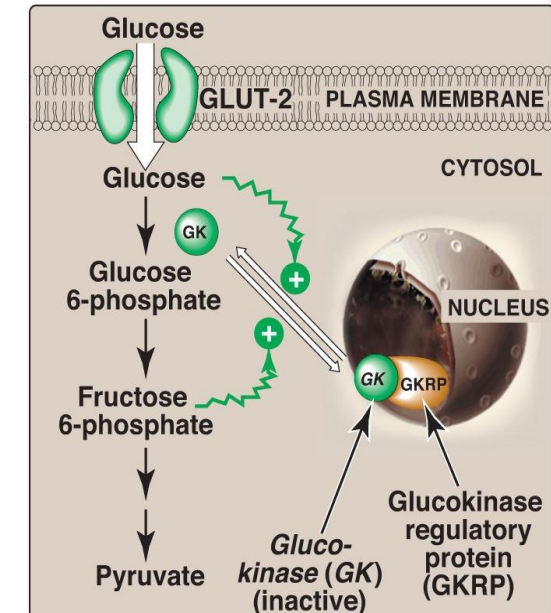


Glucokinase

- Glucokinase (liver) **high K_m ; high V_{max}**
 - More active as blood glucose level elevated (after a meal)
- No product inhibition by glucose 6-phosphate
- ‘Glucose sensor’ in liver and beta-cells of pancreas
- Enzyme sequestered in nucleus when plasma glucose level is low
- Glucokinase Mutations: **Hyperglycemia** and **Monogenic diabetes** (previously **MODY-2**)

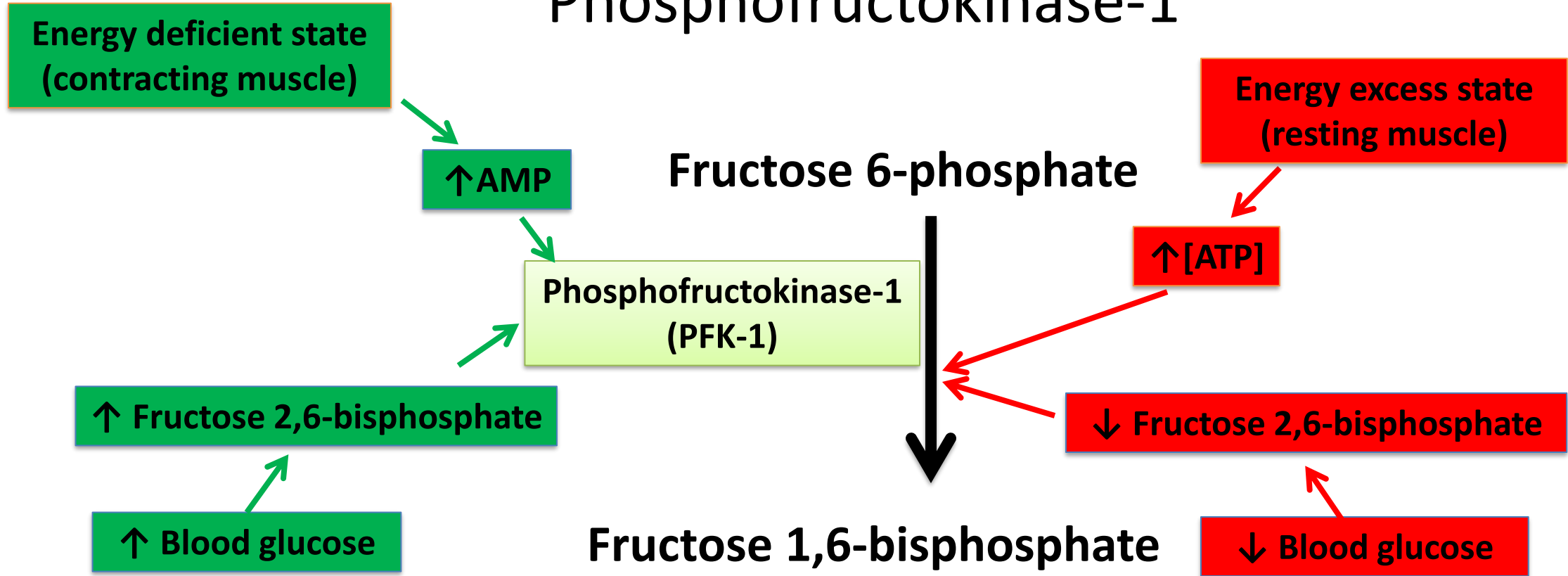


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Phosphofructokinase-1

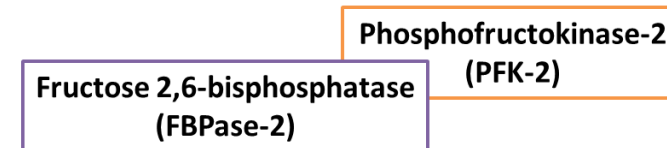


- Allosteric enzyme and committed enzyme
- Muscle: ATP (inhibits) & AMP (stimulates)
- Liver: **Fructose 2,6-bisphosphate** (stimulates) and **citrate** (inhibits)

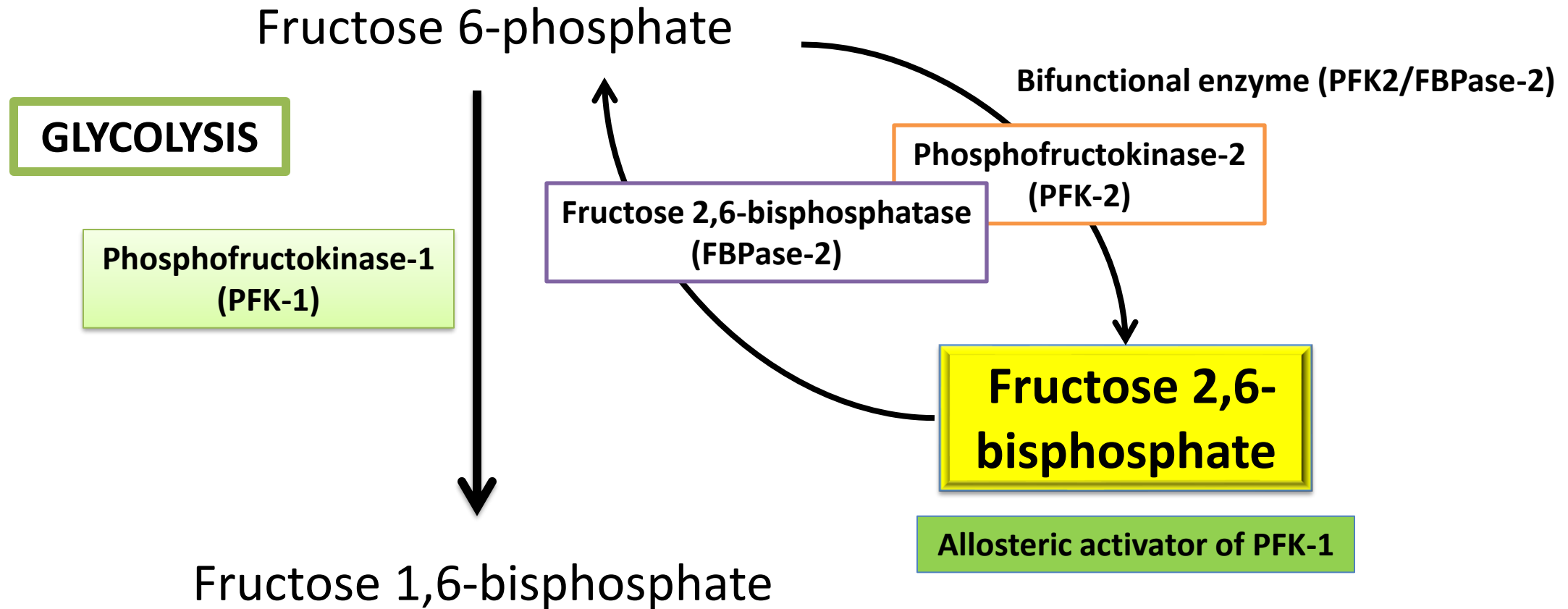
Hormonal regulation of glycolysis

Fructose 2,6-bisphosphate: formation and breakdown

- **PFK-2** forms Fructose **2,6**-bisphosphate
 - **Allosteric activator** of glycolysis and
 - Allosteric inhibitor of gluconeogenesis
- **Fructose 2,6-bisphosphate** degraded by Fructose bisphosphatase-2 (FBPase-**2**)
- PFK-2/ FBPase-2 : '**Bifunctional enzyme**'
- **Phosphorylation** by **glucagon**
 - Glucagon → activates adenylate cyclase → Increased cAMP → Activates protein kinase A → Phosphorylates key enzymes
- **Dephosphorylation** by **insulin**
 - Insulin → Activates phosphoprotein phosphatase → Dephosphorylates key enzymes
- **Fructose 1,6-bisphosphate**: intermediate of glycolysis and gluconeogenesis



Fructose 2,6-bisphosphate: formation and breakdown

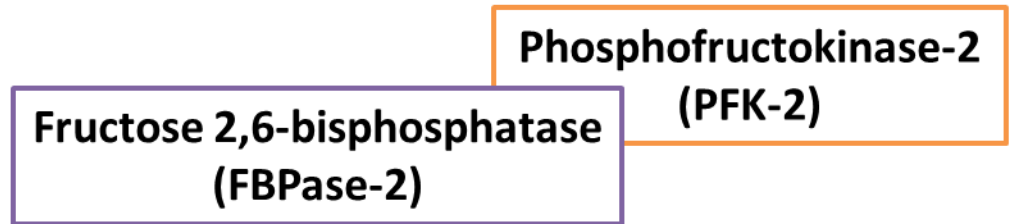


Recap and review: Fructose 2,6-bisP:
<https://sgu1.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=0130dc1b-89a1-4b7b-aac3-b3d301255a59>

Hormonal Regulation of Bifunctional enzyme and role of Fructose 2,6-bisphosphate

- **Insulin (high blood glucose level)**

- Bifunctional enzyme: Dephosphorylation (Insulin activates phosphoprotein phosphatase-1)
- Active **PFK-2**
- **Increases [Fructose 2,6-bisphosphate]**
- Allosterically activates **PFK-1 (glycolysis)**

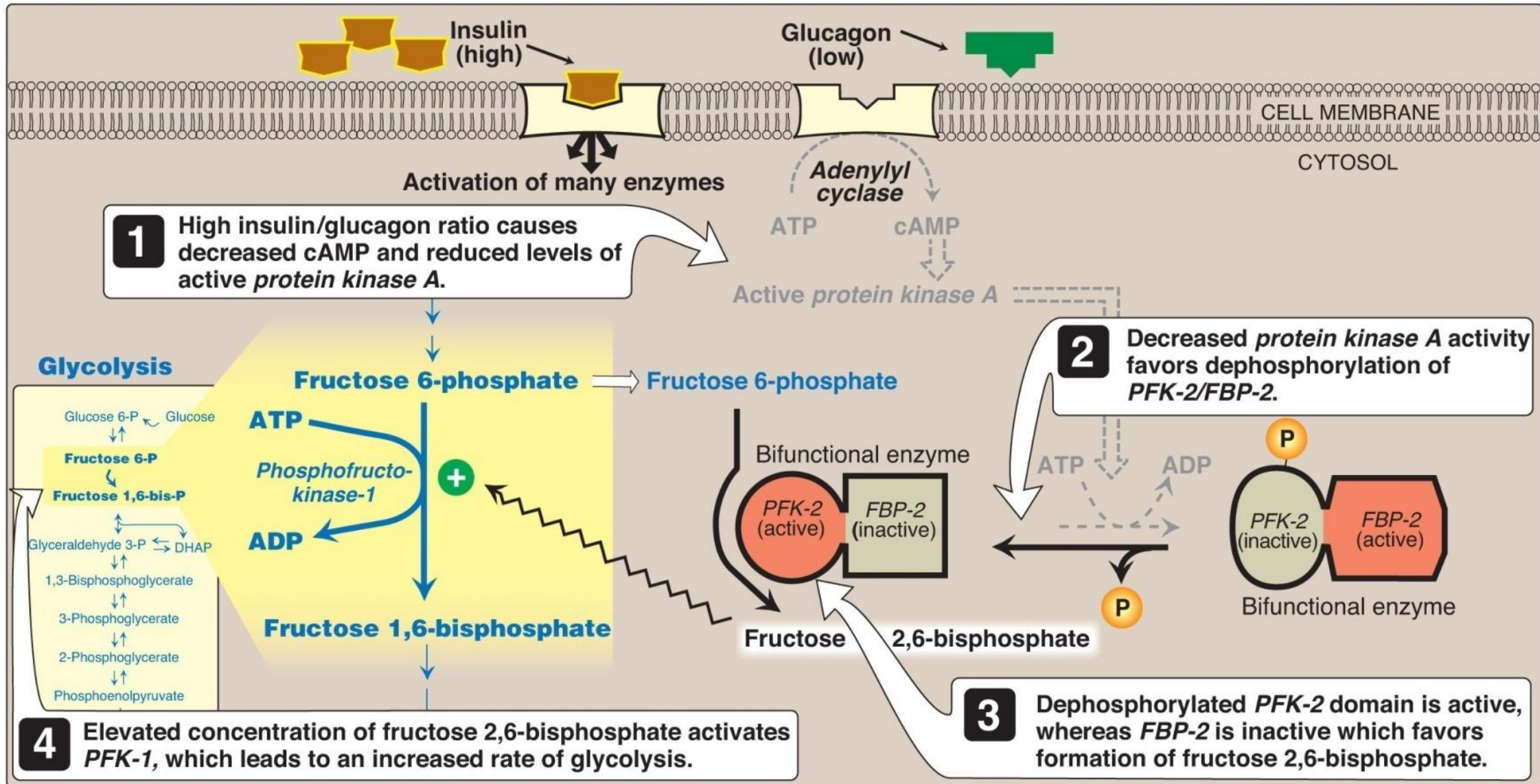


- **Glucagon (low blood glucose level)**

- Bifunctional enzyme: Phosphorylation (Active Protein kinase A)
- **FBPase-2** active
- **Decreases [Fructose 2,6-bisphosphate]**
- Activates **Fructose 1,6-bisphosphatase (gluconeogenesis)**

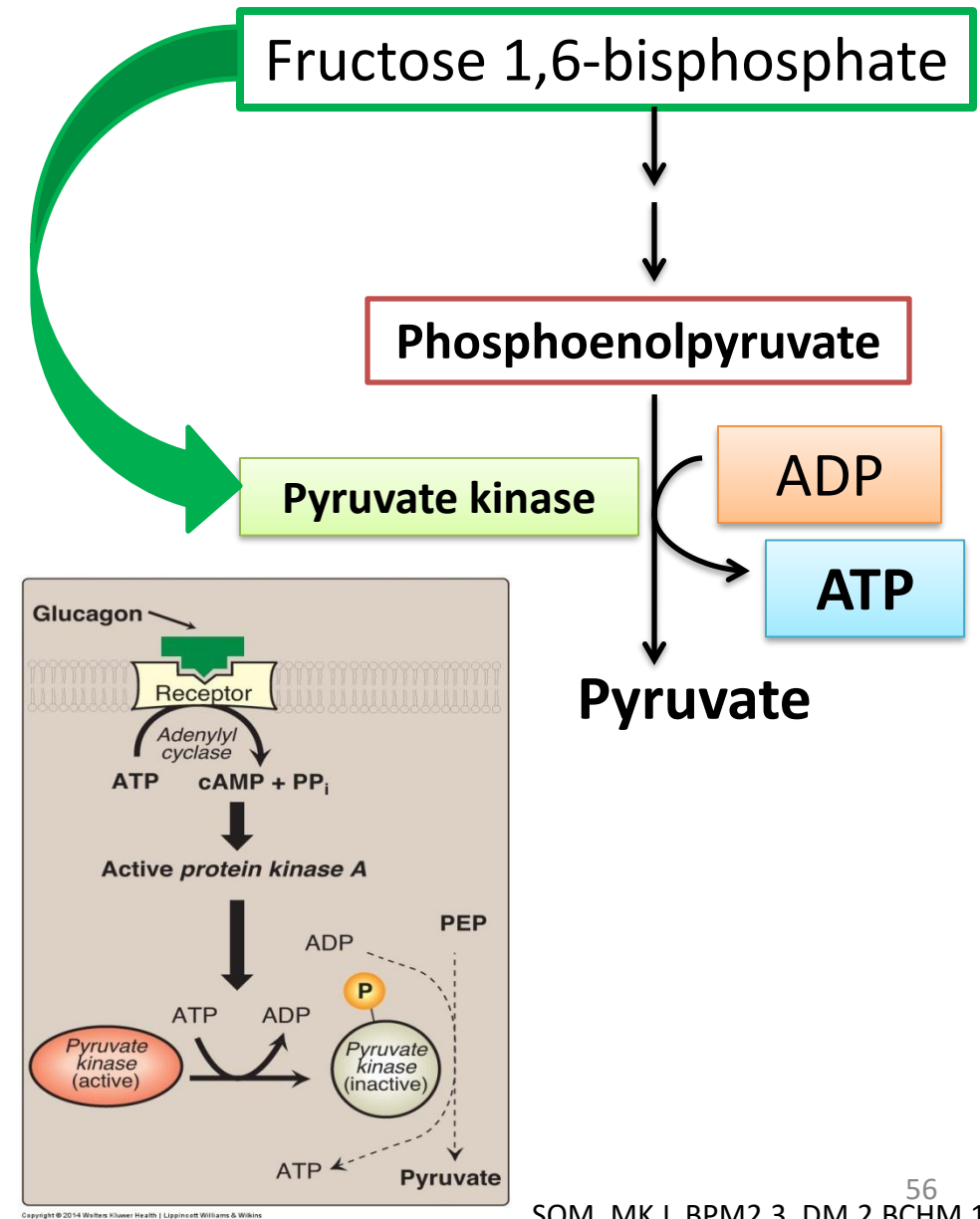
- Also refer to gluconeogenesis; regulation of carbohydrate metabolism

High insulin: glucagon ratio: PFK-1 Regulation: Role of Fructose 2,6-bisphosphate



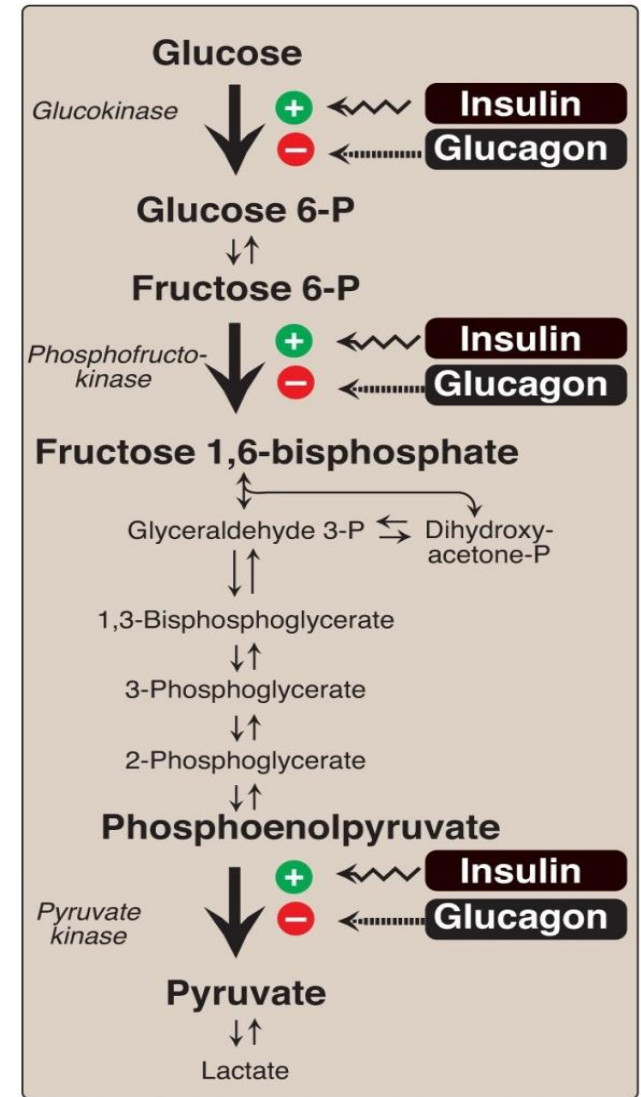
Pyruvate kinase Regulation

- Feedforward activator: **Fructose 1,6-bisphosphate**
- **Phosphorylation** (glucagon) **decreases** activity
- Low blood glucose: Liver glycolysis less active, and gluconeogenesis active.
- **Alanine** (substrate for gluconeogenesis) **inhibits pyruvate kinase**



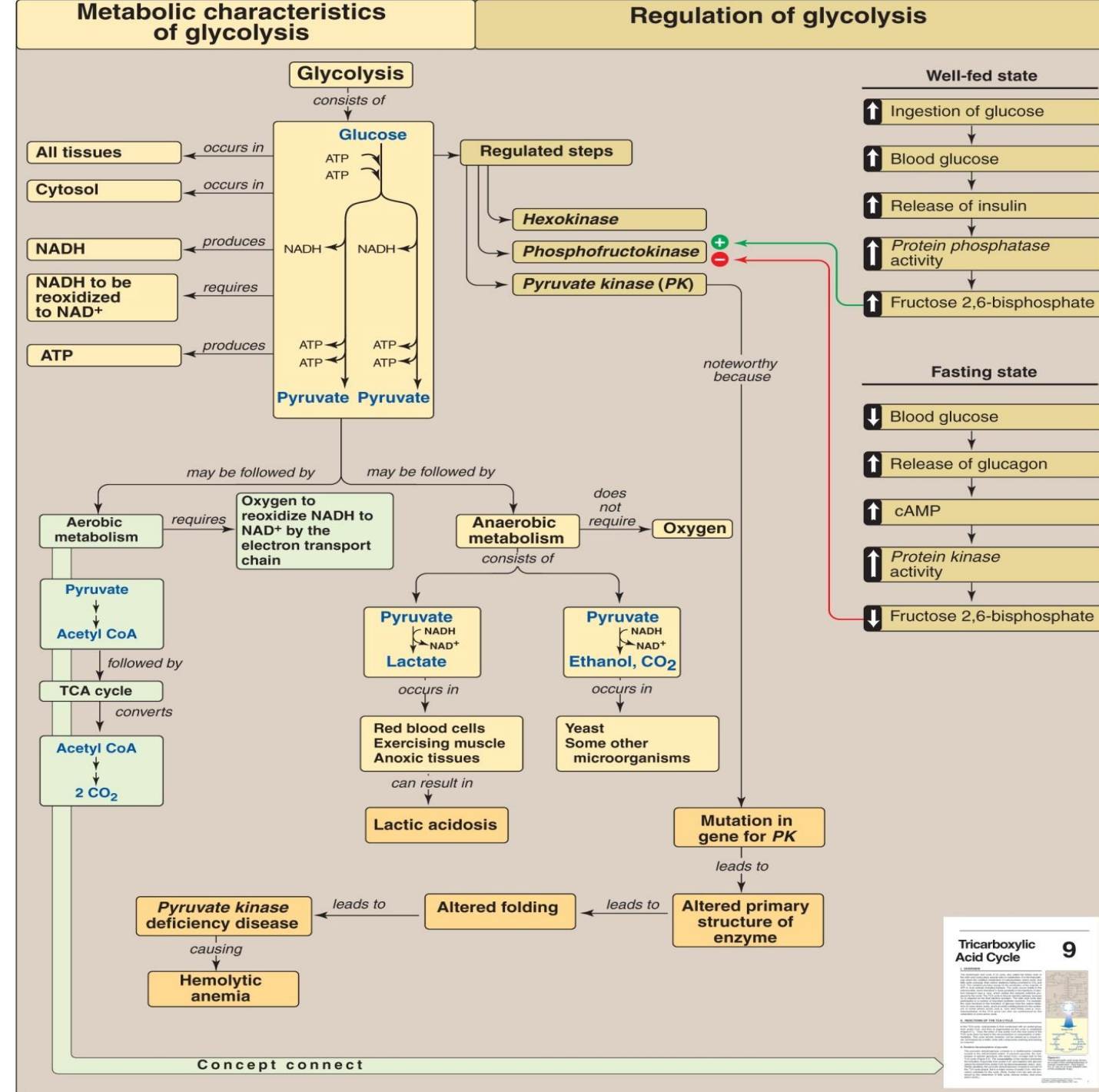
Long term regulation

- **High carbohydrate diet and insulin: induce** glycolytic enzymes (Increased synthesis increases concentration of enzymes)
 - Low levels of insulin (diabetes mellitus) represses glycolytic enzymes
- **Glucagon and fasting repress** glycolytic enzymes



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Concept Map Fig 8.25



Required post-reading and Objectives: DLA: Structure and function - Carbohydrates

- After DM Lecture 01

SOM. MK.I. BPM2.3. DM.2.BCHM.1099	Outline classification of monosaccharides based on number of carbon atoms and functional group (aldoses and ketoses)
SOM. MK.I. BPM2.3. DM.2.BCHM.1100	Differentiate between: Reducing and non-reducing sugars
SOM. MK.I. BPM2.3. DM.2.BCHM.1101	Define epimer using galactose as an example
SOM. MK.I. BPM2.3. DM.2.BCHM.1102	Compare and contrast the bonds present in: Monosaccharides (glucose, galactose, fructose); Disaccharides (maltose, sucrose, lactose); Polysaccharides (starch, glycogen and cellulose)
SOM. MK.I. BPM2.3. DM.2.BCHM.1103	Identify significance of sugar alcohols (sorbitol, galactitol) and sugar acids (glucuronic acid)
SOM. MK.I. BPM2.3. DM.2.BCHM.1104	Compare sucrose with High Fructose Corn Syrup (HFCS)
SOM. MK.I. BPM2.3. DM.2.BCHM.1105	Review clinical significance of detecting reducing sugars in urine in diabetes mellitus and inherited disorders of fructose and galactose metabolism

Required post-reading and Objectives: DLA: Overview of metabolism in the fed fasting state

- After DM Lecture 01

SOM. MK.III. BPM2.3. DM.2.BCHM. 1106	Outline the metabolic changes that occur during the feed/fast cycle.
SOM. MK.III. BPM2.3. DM.2.BCHM. 1107	Discuss pathways that are active/inactive in each major organ/tissue during the feed/fast cycle and their regulation and coordination.